

Development of Multivendor Ecommerce Application for Cloudly Infotech Limited

Md. Mahadi Hasan

ID# 20103206

A Practicum in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT – International University of Business Agriculture and Technology

Fall 2023

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The practicum has been examined and approved,

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Department of Computer Science and Engineering
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IUBAT – International University of Business Agriculture and Technology

Fall 2023

Letter of Transmittal

17 December 2023

The Chair

Practicum Defense Committee

Department of Computer Science and Engineering

IUBAT–International University of Business Agriculture and Technology

4 Embankment Drive Road, Sector 10, Uttara Model Town

Dhaka 1230, Bangladesh.

Subject: Letter of Transmittal.

Dear Sir, with due regards, I am Md. Mahadi Hasan, I'd want to approach you that it would be an exceptional open door and a great joy for me to provide you this report, which is titled “Development of Multivendor Ecommerce Application for Cloudly InfoTech Limited” for the fulfillment of my practicum course. I did my absolute best to gather as much relevant data for the report as I could. I gained a lot of practical experience while writing the report, which will be very beneficial to me in the future. I now have more practical knowledge as a result. If you need any further explanation, I'll be happy to provide it.

I appreciate you giving me the chance to write a report on the aforementioned subject.

Yours sincerely,

Md. Mahadi Hasan
ID#20103206

Organization Certificate



CERTIFICATION

Date: 13 December 2023

This is to certify that Md. Mahadi Hasan, who completed B.Sc. in Computer Science from International University of Business, Agriculture and Technology, Dhaka, did his internship from 01 October 2023 and expected end date is 31st December 2023 (40 hrs/week).

As an Software Developer (Intern), his major responsibilities are-

- UI Development
- Feature Implementation
- Code debugging and Testing.
- Performance optimization
- Collaboration and code reviews.

Thank You

A handwritten signature in black ink, appearing to read "Ishmam".

Cloudly Infotech Limited

Ishmam Mahdy
Manager (HR, Admin & Accounts)
Cloudly Infotech Limited

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Student's Declaration

I am Md. Mahadi Hasan, bearing ID#20103206, student of BCSE - Bachelor of Computer Science and Engineering program, under the College of Engineering and Technology (CEAT) of IUBAT - International University of Business Agriculture and Technology declaring that, this report on the topic of “Development of Multivendor Ecommerce Application for Cloudly Infotech Limited” has been prepared for the fulfillment of the internship CSC 490, Practicum as well as the partial requirement of BCSE-Bachelor of Computer Science and Engineering degree.

I am the author of the report and project on the "Multivendor Ecommerce Application for Cloudly Infotech Limited". This project's modules and processes were created following thorough research on the internet.

There are no other uses, prizes, or presentations for which it has been prepared.

Md. Mahadi Hasan
ID#20103206

Supervisor's Certification

For the purpose of certifying that the practicum report on “Multivendor Ecommerce Application for Cloudly Infotech Limited” has been executed by Md. Mahadi Hasan bearing ID#20103206 of IUBAT – International University of Business Agriculture and Technology as a partial completion of the obligation of practicum defense course. The report, which was created with my assistance, serves as a summary of the successfully completed task. No portions of this report have, to the best of my knowledge and according to his assertion, been submitted anywhere for any degree, diploma, or certification. He can submit the report at this point. I hope he succeeds in everything he does moving forward.

Md Saidur Rahman

Supervisor and Assistant Professor

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Abstract

The Multi-Vendor E-commerce Platform is a comprehensive web application designed to facilitate seamless interactions between customers, vendors, transporters, delivery personnel, and administrators within an expansive online marketplace. Leveraging the power of the MERN (MongoDB, Express.js, React.js, Node.js) stack, this platform provides a dynamic and user-friendly interface for a diverse range of users. Customers are empowered with an array of functionalities, including the ability to browse and purchase products, track orders, provide ratings and reviews, request refunds, update their profiles, engage in chat communications with vendors, and make secure payments. Additionally, the system ensures a streamlined login and registration process for user convenience and security. Vendors enjoy a robust suite of tools enabling them to create, update, and manage their individual shops, along with the flexibility to add, modify, or remove products from their inventory. They can further enrich customer engagement through events such as discounts and special offers, manage order statuses, accept or reject refund requests, and efficiently handle financial transactions, including balance withdrawals. Facilitating the logistics side of the operation, transporters are responsible for assigning orders to delivery personnel. This assignment chain, originating from vendors and managed by transporters, ensures efficient and timely order deliveries to customers. The system administrator plays a pivotal role in maintaining the platform's integrity and functionality. They have the authority to create new transporter and deliveryman accounts, monitor the system's performance, and uphold its security standards by managing user access and permissions. Overall, this multifaceted ecosystem brings together various stakeholders in a unified digital marketplace.

Acknowledgments

In the name of Almighty, who is the most merciful and the most graceful, it's my pleasure to take this occasion to thank a few people, who have assisted, encouraged, directed and supported me during my practicum program. Firstly, I want to thank my family, who has endowed their immeasurable-innumerable support and encouragement to attain this exquisite event of my life. I am very appreciative to Prof. Dr. Utpal Kanti Das, Chair of Department of Computer Science and Engineering, IUBAT- International University of Business Agriculture and Technology for his unrelenting direction and sustain throughout the semester. I would like to pay my gratitude to my practicum supervisor Md Saidur Rahman, Assistant Professor of Computer Science and Engineering Department, who has given me the opportunity to make such a report for not only in this semester but also throughout my education life at IUBAT by giving her valuable suggestions and advices at any time, at any situation. I would be able to make this report effectively and properly only for her right direction.

Sincerely,

Md. Mahadi Hasan

ID#20103206

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Chapter 1

Introduction

1.1 Introduction

In the ever-evolving landscape of digital commerce, the Multi-Vendor E-commerce Platform stands as an innovative beacon, transcending traditional shopping paradigms by fostering a harmonious ecosystem that converges customers, vendors, transporters, delivery personnel, and administrators into a dynamic and interconnected marketplace.

At its core, this project redefines the online shopping experience by not merely facilitating transactions, but by orchestrating a symphony of interactions, empowering each participant with specialized tools and functionalities. Rooted in the powerful MERN stack—MongoDB, Express.js, React.js, and Node.js—the platform embodies sophistication and efficiency, elevating the art of e-commerce to new heights.

For customers, it's an immersive journey of exploration and convenience. From browsing a diverse array of products to seamlessly completing transactions, tracking orders, providing feedback, and engaging in real-time conversations with vendors, the platform prioritizes user experience at every turn. A robust yet intuitive interface simplifies tasks while ensuring a secure and personalized shopping environment.

Vendors, the heartbeat of this digital marketplace, find themselves equipped with a comprehensive toolkit to curate and manage their individual shops with unparalleled ease. Product creation, updates, and inventory management are seamlessly integrated with functionalities such as event-based promotions, order status management, and swift financial transactions, fostering an environment conducive to fostering customer loyalty and satisfaction.

In the realm of logistics, transporters and delivery personnel synergize to ensure the seamless fulfillment of orders. Their pivotal role in the supply chain ensures that the connection between vendor and customer transcends digital boundaries, culminating in timely and efficient deliveries. Not to be overlooked is the custodianship of the system administrator, entrusted with maintaining the platform's integrity and security. This role extends beyond user management, encompassing system monitoring, new user onboarding, and safeguarding the platform against potential threats, ensuring a smooth and secure digital environment for all stakeholders. This project isn't just about facilitating transactions; it's a testament to collaboration, innovation, and inclusivity within the digital commerce landscape. By

seamlessly interconnecting various stakeholders, it creates an ecosystem where transactions transcend mere exchanges, fostering relationships and experiences that redefine the essence of online commerce. Welcome to the Multi-Vendor E-commerce Platform—a visionary convergence of technology, collaboration, and limitless possibilities.

1.2 Background Study

This project emerged from an analysis of the shifting e-commerce landscape, recognizing the growing demand for multi-vendor platforms. Research highlighted the limitations of single-vendor systems and the potential in enhancing customer-vendor interactions, optimizing logistics, and empowering administrators. The decision to adopt the MERN stack was based on its scalability and adaptability, aligning with the platform's need for a robust foundation. Insights from user behavior, market trends, and industry standards guided the integration of user-centric features and intuitive interfaces, emphasizing seamless vendor-customer interactions. This study anchors the project's development, aligning its vision with evolving user needs, technological capabilities, and emerging trends in the online commerce realm.

1.3 Objective

The project aims to create an inclusive MERN stack-based multi-vendor e-commerce platform. For customers, it seeks to offer a seamless experience with secure registration, shopping convenience from various vendors, order tracking, efficient refund management, interactive product reviews, profile customization, diverse payment options, and direct communication channels. Vendors will have tools for comprehensive shop management, event creation, order handling, earnings withdrawal, and customer engagement. Transporters and delivery personnel will streamline order assignments for prompt delivery. The system administrator will oversee system health, manage user accounts, ensure security, and resolve issues to maintain a reliable and efficient platform.

1.4 Features

1.4.1 Customer Features:

1. **Registration and Authentication:** Implement secure registration and login functionalities for users.
2. **Shopping Capabilities:** Develop an intuitive interface for browsing products from various vendors and facilitating purchases.
3. **Order Tracking:** Enable customers to track their orders in real-time within the platform.
4. **Refund Management:** Provide a streamlined process for managing refunds, including submission and tracking of refund requests.
5. **Rating and Reviews:** Allow customers to rate and review products, enhancing transparency and aiding other shoppers.
6. **Profile Management:** Enable users to easily update and manage their profiles and preferences.
7. **Payment Options:** Integrate multiple payment gateways to ensure flexible and secure transactions.
8. **Customer-Vendor Communication:** Implement a chat feature for direct communication between customers and vendors for inquiries or support.

1.4.2 Seller Features:

1. **Shop Management:** Enable vendors to create, update, and delete their shops and product listings.
2. **Event Creation:** Allow vendors to create promotional events such as discounts and special offers.

3. **Order Management:** Provide tools for vendors to manage order statuses, including acceptance, rejection, and handling refund requests.
4. **Earnings Withdrawal:** Facilitate the withdrawal of earnings for vendors from their sales on the platform.
5. **Customer Interaction:** Implement communication channels between vendors and customers for efficient support and engagement.

1.4.3 Transporter and Delivery Personnel Features:

1. **Order Assignment:** Assign transporters to manage the logistics of order allocation to delivery personnel.
2. **Efficient Delivery Process:** Streamline the assignment and delivery process for prompt and efficient order fulfillment.

1.4.4 System Administrator Features:

1. **System Monitoring:** Monitor the platform's performance and functionality to ensure smooth operation.
2. **User Management:** Create and manage accounts for transporters, delivery personnel, and other users as needed.
3. **Security Oversight:** Maintain system integrity and security by monitoring user and shop activities, intervening when necessary.
4. **Issue Resolution:** Have the authority to resolve platform-related issues and maintain overall system health.
5. **Account Management:** Manage user and shop accounts, ensuring compliance and reliability across the platform.

1.5 Proposed System Benefits

The proposed system offers a multitude of benefits across user roles:

Enhanced Shopping Experience:

- Customers benefit from a user-friendly interface, diverse product options, and seamless transactions, leading to a satisfying shopping journey.
- Vendors can efficiently manage their shops, engage with customers, and drive sales through targeted events and improved communication.

Efficiency and Transparency:

- The integrated order tracking system ensures transparency for customers while optimizing order management for vendors and delivery personnel.
- Refund management tools streamline processes, ensuring quick and transparent resolution for both customers and vendors.

Improved Communication and Support:

- Real-time chat functionalities facilitate direct and instant communication between customers and vendors, fostering better support and resolving queries promptly.

Empowered Administration:

- The system administrator gains comprehensive oversight, ensuring the platform's smooth operation, resolving issues efficiently, and maintaining security and reliability.
- Transporters and delivery personnel experience streamlined workflows, optimizing order assignment and delivery processes, enhancing efficiency.

Financial Control and Growth:

- Vendors benefit from event creation tools, enabling them to attract customers with promotions and discounts, while also efficiently managing order statuses and earnings.

Scalability and Adaptability:

- The system's architecture allows for scalability, accommodating an increasing user base and adapting to evolving needs in the e-commerce landscape.

Overall, the proposed system facilitates a seamless and efficient multi-vendor e-commerce ecosystem, benefiting customers, vendors, administrators, and logistical personnel alike, while fostering growth, transparency, and reliability within the platform.

1.6 Process Model

The incremental process model is proposed for the development of the web-based Development of Multivendor Ecommerce Application for Cloudly InfoTech Limited. This model is suitable when the software requirements are well-defined, but there is a need for quick delivery of the software to meet user demands.

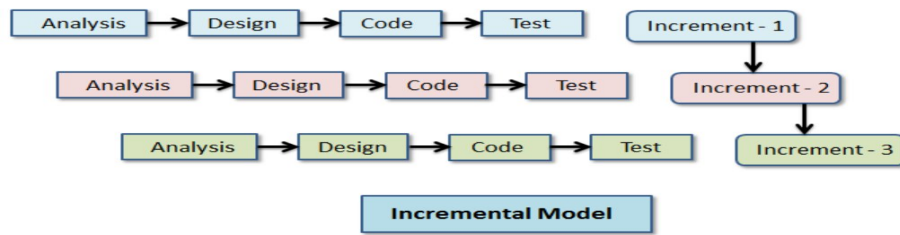


Figure 1.1: Incremental Process Model

1.7 Reason for choosing incremental process model:

The decision to choose the incremental process model for the development of the web-based Development of Multivendor Ecommerce Application for Cloudly InfoTech Limited is based on the following reasons:

- **Adaptability to Changing Requirements:** Incremental models allow for flexibility in accommodating evolving project requirements over time. As this project involves various stakeholders with diverse needs, this model allows for iterative development, incorporating changes as necessary.

- **Iterative Development:** The incremental model facilitates phased development where each phase focuses on specific functionalities. This approach ensures thorough testing and validation of each component before integration, reducing the risk of errors in the final product.
- **Early Delivery of Core Functionalities:** By breaking down the project into manageable increments, essential functionalities can be developed and delivered early in the process. This allows for a quicker time-to-market for key features, benefiting stakeholders and end-users.
- **Continuous Feedback and Improvement:** The model allows stakeholders to provide feedback at various stages of development. This continuous feedback loop ensures that the end product aligns closely with stakeholder expectations and needs, enhancing overall satisfaction.
- **Risk Management and Containment:** Incremental development mitigates project risks by addressing them in smaller, manageable segments. Issues and challenges can be identified early in the process, minimizing their impact on subsequent phases.
- **Enhanced Quality Assurance:** Each increment undergoes rigorous testing and validation, ensuring higher quality in each functional component. This approach reduces the likelihood of major defects in the final product.
- **Better Resource Utilization:** The incremental model optimizes resource allocation by focusing on specific functionalities in each phase. This targeted approach ensures efficient utilization of resources, minimizing wastage.
- **Clear Milestones and Progress Tracking:** With clearly defined increments, the project team can set achievable milestones for each phase. This allows for better tracking of progress, making it easier to monitor and manage the project's development lifecycle.

The choice of the incremental process model for this project is founded on its ability to offer flexibility, iterative development, continuous improvement, risk mitigation, and efficient resource utilization throughout the project's lifecycle

1.8 Feasibility study

A feasibility study is conducted to determine whether the proposed solution, in this case, the web-based point of sale system, is feasible and achievable for the organization. The feasibility study involves investigating three main areas:

- 📦 Technical feasibility
- 📦 Economical feasibility
- 📦 Operational feasibility

Technical Feasibility:

Employing the MERN stack coupled with Socket.io, Tailwind CSS, and Material UI indicates strong technical feasibility due to their adaptability and robust functionalities. Socket.io facilitates real-time, bidirectional communication between users, enhancing user interaction without relying on external third-party services. Tailwind CSS and Material UI frameworks offer comprehensive styling and design elements, streamlining UI development without additional dependencies. The integration of essential functionalities such as order tracking, payment gateways, and map APIs for delivery tracking within this tech stack is viable, albeit requiring meticulous implementation to ensure seamless operation and security.

Economical Feasibility:

Accurate estimation of development costs, encompassing resources like developers, design tools, and infrastructure, is critical for financial planning. Ongoing expenses for maintenance, updates, and support should be factored into the economic feasibility analysis. Evaluating market demand, competitive landscape, and revenue models such as commission structures from sales or subscription fees is necessary to sustain operational costs and ensure profitability. A thorough Return on Investment (ROI) analysis considering the projected revenue against development and operational expenses will validate the project's economic viability.

Operational Feasibility:

Prioritizing an intuitive and responsive user experience (UX) for customers, vendors, transporters, and administrators is essential. Simplifying vendor management processes through Tailwind CSS and Material UI will streamline onboarding, product listing, and shop management. Socket.io facilitates real-time communication, enhancing order processing and customer support without relying on external services. Robust order management features and efficient handling of refunds or returns within this framework will contribute to operational efficiency. Comprehensive administrative controls will empower administrators to manage users, shops, and system operations effectively.

Conducting a feasibility study enables assessing the viability of a project comprehensively. It provides crucial insights into potential risks, helping in informed decision-making and resource allocation. By understanding the technical, economic, and operational aspects beforehand, we can minimize risks, optimize resource utilization, and increase the likelihood of successful project outcomes.

Chapter 2

Organizational Overview

2.1 Organizational Overview

Cloudly Infotech Limited stands as a dynamic and forward-thinking entity, offering an extensive array of unparalleled cloud services and cutting-edge technological solutions. Our commitment extends beyond conventional cloud services, encompassing a diverse portfolio that includes software development, innovative web solutions, mobile application development, graphic design, and multimedia services.

Distinguished as a renowned cloud service provider, Cloudly Infotech Limited is not only a pioneer in revolutionizing businesses through cloud technologies but also in catering to the evolving needs of enterprises both domestically and globally. Our commitment to quality and innovation is evident in our multifaceted approach, addressing the diverse requirements of clients across industries.

Operating as a catalyst in the technological sphere, Cloudly Infotech Limited leverages its expertise to empower businesses to thrive in today's competitive landscape. Our mission is not just to provide exceptional services but also to create opportunities for professional growth and employment. Our team comprises highly qualified individuals, passionate about innovation and committed to leading in their respective fields.

At Cloudly Infotech Limited, client satisfaction is paramount. We offer specialized solutions tailored to the unique needs of each client, ensuring they harness the full potential of cloud technologies to achieve unprecedented success in their ventures. United by a common goal and unwavering passion, our team works cohesively to guide our clients towards prosperity in the ever-evolving digital era.

2.2 Mission:

At Cloudly Infotech Limited, our mission is to pioneer transformative cloud solutions and technological innovations that empower businesses to thrive in the digital landscape. We are dedicated to delivering exceptional services tailored to each client's unique needs, fostering growth, and creating opportunities for success in an ever-evolving technological world.

2.3 Vision:

Our vision at Cloudly Infotech Limited is to be the foremost leader in revolutionizing businesses through cutting-edge cloud technologies. We aspire to be recognized globally for our commitment to innovation, exceptional service delivery, and the creation of a dynamic ecosystem where businesses flourish by harnessing the power of the cloud.

2.4 Services

Software Solutions:

Cloudly Infotech Limited delivers bespoke software solutions tailored to address the specific needs and challenges of our clients. Our expertise spans across the complete software development lifecycle, from conceptualization and design to development, testing, and maintenance. We specialize in crafting scalable, secure, and innovative software applications, leveraging the latest technologies to drive business growth and efficiency.

DevOps Services:

Cloudly Infotech Limited offers comprehensive DevOps solutions aimed at streamlining and accelerating software development and deployment processes. Our DevOps services foster collaboration between development and operations teams, integrating automation, continuous integration/continuous deployment (CI/CD), and agile methodologies. By optimizing workflows and infrastructure, we empower businesses to achieve faster time-to-market, enhanced reliability, and improved scalability.

Cloud Integration Services:

Our Cloud Integration Services are designed to seamlessly merge diverse applications, systems, and data across cloud platforms, ensuring smooth interoperability and functionality. Cloudly Infotech Limited specializes in integrating various cloud services and applications, providing a unified ecosystem for businesses. Our expertise in cloud integration enables efficient data flow, real-time synchronization, and enhanced accessibility, enabling businesses to harness the full potential of cloud-based solutions.

These services collectively empower businesses to embrace technological advancements, optimize their operations, and achieve sustainable growth in an ever-evolving digital landscape. Cloudly Infotech Limited's commitment lies in providing holistic solutions that enable businesses to thrive and succeed in the dynamic realm of cloud computing and software development.

2.5 Organization Location

House #429(2nd Floor), Road#30, Dhaka-1206

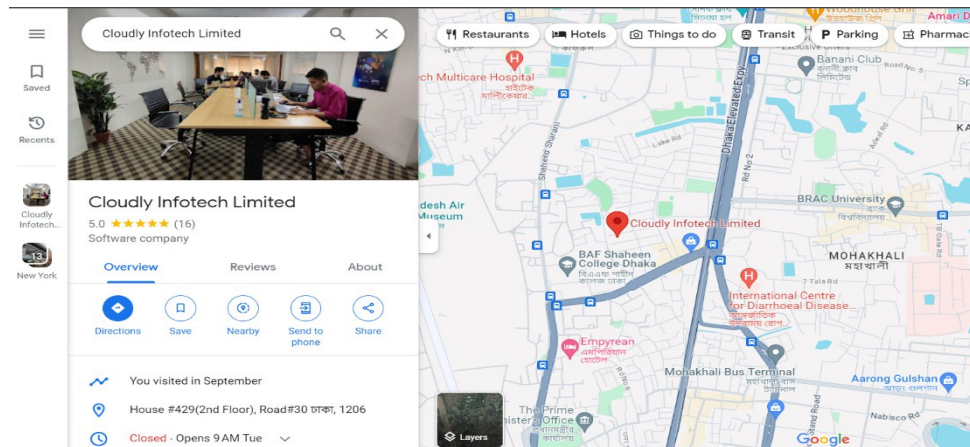


Figure 2. 1 Organization's location (Source: Google map)

2.6 My designation in this organization

During my internship at Cloudly Infotech Limited as a Software Engineer, I have had the opportunity to work under the guidance of a supportive and informative supervisor. I am grateful for the knowledge and skills I have gained through this experience. With the guidance of my supervisor, I successfully completed my project within the allotted time frame. This internship has not only provided valuable technical experience but also taught me the importance of adhering to office hours and following organizational rules. I am truly grateful for the opportunity to work with this office, as it has prepared me for the beginning of my career journey.

Chapter 3

Requirement Engineering

3.1 Requirement Engineering

Requirement engineering is a systematic process of identifying, documenting, and managing the requirements of a system. It involves understanding the needs and expectations of stakeholders, analyzing those needs, and translating them into clear and unambiguous requirements that can guide the design, development, and implementation of the system.

There are several types of requirements that are typically addressed in requirement engineering:

- User requirements
- System requirements
- Functional requirements
- Non-functional requirements
- Hardware Requirements
- Software Requirements

3.2 User Requirement:

Customer:

- Customer should be able to login/register.
- Customer should be able to view product.
- Customer should be able to search product.
- Customer should be able to see the details of the product.
- Customer should be able to manage the cart items.
- Customer should be able to add product in favorite list.
- Customer should be able to manage their profile
- Customer should be able to place order of an available product.
- Customer should be able to make payment.
- Customer should be able to manage addresses.

- Customer should be able to track order.
- Customer should be able to give reviews and ratings.
- Customer should be able to view shop page.
- Customer should be able to contact with the Shop/Seller.
- Customer should be able to refund the product.

Shop/Seller:

- Seller should be able to login/register
- Seller should be able to manage products.
- Seller should be able to manage events.
- Seller should be able manage discount coupons.
- Seller should be able manage orders.
- Seller should be able to track and update order status.
- Seller should be able to manage refunds.
- Seller should be able to manage shop profile
- Seller should be able to withdraw their money.

Transporter:

- Transporter should be able to login/register.
- Transporters should be able to view assigned orders.
- Transporter should be able to update order status
- Transporter should be able to assign order to deliveryman.
- Transporters should be able to track orders.

Delivery Man:

- Deliverymen should be able to login/register.
- Deliverymen should be able to view assigned orders.
- Deliveryman should be able to update order status.

Admin:

- Admin should be able to manage customers.
- Admin should be able to manage shops.
- Admin should be able to monitor products.
- Admin should be able to manage withdrawal requests.
- Admin should be able to manage transporter and delivery man

3.3 System Requirement:

Customer:

- Customer should be able to login/register
 - After coming to the landing page on the right side there will be a person icon when customer will click on it then customer will be redirected to sign in page. If customer is already registered, then they can provide credentials and click on login button that will redirect customer to the profile page. If credentials didn't match or customer didn't find, then error message will be displayed otherwise successful login message will be displayed
 - If a customer is not registered, then below the login button there will be a button for signup. If a customer clicks on it then they will be redirected to the signup page and then after giving credentials they can register themselves. After successful registration they will be redirected to the login page rest of the login process is the same as previously mentioned. Here also if customer provided email match with the database error message will pop up or any other error occurs that will also be displayed otherwise successful registration message will be displayed
- Customers should be able to view products.

- Customers should be able to view all the products. If he clicks on the products menu in the navbar also to view products, customers don't need to login.
- Customers should be able to search for products.
 - There should be a search bar for customer.
 - Customer can enter text there to search their product and also a search icon by clicking that customer will only show the searched product.
 - There should be a category option upon selecting any specific category customer will only see products from that category only.
- Customer should be able to see the details of the product:
 - Clicking on a product displays its detailed information.
 - Information includes product description, images, price, availability, reviews, and related products.
 - Options for different variants, sizes, colors, etc.
- Customer should be able to manage the cart items:
 - An interface to view, add, remove, and update items in the shopping cart.
 - Ability to view total cost, apply discounts or promo codes, and proceed to checkout.
- Customer should be able to add product to a favorite list:
 - Capability to add products to a personal wish list or favorites list for future reference.
 - Option to manage and remove items from the favorites list.
- Customer should be able to manage their profile:
 - Access to edit personal information like name, address, contact details, and password.
 - Option to update preferences, subscription settings, and communication preferences.

- Customer should be able to place an order of an available product:
 - A streamlined checkout process to confirm and place orders securely.
 - Confirmation notifications upon successful order placement.
- Customer should be able to make payment:
 - Secure payment gateways allowing various payment methods like credit/debit cards, digital wallets, or other relevant options.
- Customer should be able to manage addresses:
 - Ability to add, edit, or remove multiple shipping addresses.
 - Option to set a default shipping address for convenience.
- Customer should be able to track orders:
 - Real-time order tracking functionality to monitor order status and estimated delivery times.
 - Order history for reference and tracking updates.
- Customer should be able to give reviews and ratings:
 - Review and Rating System:
 - Capability to rate products and provide detailed reviews based on experiences.
 - Reviews visible to other customers for informed decision-making.
- Customer should be able to view the shop page:
 - Access to a dedicated shop page showcasing all products from a particular seller/shop.
 - Easy navigation to shop-specific categories, featured items, and promotions.
- Customer should be able to contact with the Shop/Seller:
 - Contact options like chat to connect with the seller for inquiries, support, or assistance.
- Customer should be able to refund the product:

- Clear guidelines and procedures for initiating product returns or refunds.
- Tracking of refund status and notifications upon completion.

Shop/Seller:

- Seller should be able to login/register
 - Similar to the customer login, with dedicated access for sellers. Secure authentication requiring unique seller credentials.
 - Registration portal specifically designed for sellers.
 - Fields might include business name, tax ID, contact information, and bank details for transactions.
- Sellers should be able to manage products.
 - Seller dashboard allowing the addition, editing, and removal of products. Options for managing product details, images, pricing, inventory, and categorization.
- Seller should be able to manage events.
 - Interface to create, manage, and promote events or sales campaigns. Options for scheduling, modifying, and tracking event performances.
- Sellers should be able to manage discount coupons.
 - Capability to create discount coupons with specific parameters like discount percentage, validity, and product applicability.
 - Management interface for editing, activating, or deactivating coupons
- Sellers should be able to manage orders.
 - Dashboard displaying orders received, with details such as customer information, products ordered, and order status.
 - Ability to process, update, and fulfill orders efficiently.
- Sellers should be able to track and update order status.

- Real-time tracking of order status from processing to delivery.
- Options to update and communicate status changes to customers.
- Sellers should be able to manage refunds.
 - Interface to handle customer refund requests.
 - Guidelines and procedures for initiating and processing refunds.
- Sellers should be able to manage profiles.
 - Tools to manage shop information, including description, policies, contact details, and branding elements.
 - Capability to update and maintain a professional shop profile.
- Sellers should be able to withdraw their earnings.
 - Option for seller to add bank account for withdraw request.
 - Option to select amount of withdrawal.
 - Notify the seller after the withdrawal request is sent.
 - Confirmation email of the approval of withdraw request.

Transporter:

- Transporters should be able to login/register.
 - The system shall provide a secure login and registration mechanism for transporters.
 - Transporters must have unique credentials to access the platform securely.
- Transporter should be able to view assigned orders:
 - Upon login, the system shall display a comprehensive dashboard for transporters showcasing assigned orders.
 - Transporters should have immediate access to order details including pickup/delivery locations, items, and deadlines.

- Transporter should be able to update order status:
 - Transporters shall have the capability to update the status of assigned orders (e.g., picked up, in transit, delivered).
 - Status updates should be time stamped and logged for historical reference.
- Assign Orders to Deliveryman:
 - The system shall provide a feature for transporters to assign specific orders to designated delivery personnel.
 - Transporters must be able to select available deliverymen and allocate orders efficiently.
- Transporter should be able to track order
 - Transporters should have access to a real-time order tracking mechanism.
 - The system shall display an interface enabling transporters to track orders in transit, providing updates on order location and status.

Deliveryman:

- Deliveryman should be able to login/register.
 - The system shall provide a secure login and registration mechanism for transporters.
 - Transporters must have unique credentials to access the platform securely.
- Deliveryman should be able to view assigned orders:
 - Upon login, the system shall display a comprehensive dashboard for transporters showcasing assigned orders.
 - Transporters should have immediate access to order details including pickup/delivery locations, items, and deadlines.
- Deliveryman should be able to update order status:
 - Transporters shall have the capability to update the status of assigned orders (e.g. Delivered).

- Status updates should be timestamped and logged for historical reference.

Admin:

- Admin should be able to manage customers
 - Implement secure authentication for admin access.
 - Access control mechanisms to ensure restricted access to authorized functionalities.
- Admin should be able to manage customers
 - Interface for viewing, modifying, and removing customer details.
 - Tools to manage customer accounts, including updates to contact information and order history.
- Admin should be able to manage shops
 - Dashboard for admin to oversee shops, including creation, editing, and removal.
 - Features to monitor shop performance, sales analytics, and reviews/ratings.
- Admin should be able to monitor products
 - Dashboard to monitor product listings, including details, images, pricing, inventory levels, and categories.
 - Tools for admin to modify, remove, or approve product listings submitted by sellers.
- Admin should be able to manage withdrawal
 - System for handling withdrawal requests, allowing admin to review and approve/disapprove withdrawal requests from sellers.
 - Tracking mechanism to record and monitor transactional activities related to withdrawals.
- Admin should be able to manage Transporter and Delivery Personnel Management:
 - Interface to manage transporter and delivery personnel details, including onboarding, performance tracking, and assignments.

- Tools to assign, track, and manage deliveries, ensuring efficiency and reliability.
-

3.4 Functional Requirements

Customer:

Login: The customer should be able to log into the system.

View Product: Customer should be able to search and view product

Manage Cart: The customer should be able to manage the cart items.

Order product: The customer should be able to order products.

Refund: Customer should be able to return product and get refund.

Seller:

Login: The Seller should be able to log into the system.

Manage Products: Seller should be able to manage the products.

Manage Orders: Seller should be able to manage orders

Manage Events: Seller should be able to manage the events.

Withdraw Earnings: Seller should be able to withdraw earnings.

Admin:

Login: The Admin should be able to login.

Manage Users: Admin should be able to manage users.

Manage Withdrawal: Admin should be able to manage withdrawal.

3.5 Hardware Requirements:

The provided hardware requirements are suitable for running the system smoothly and comfortably. However, please note that these specifications may vary depending on the specific software and system requirements. Here are the hardware requirements:

- 1 x Intel core i3 Processor.
- 4 GB of DDR3 RAM; and a 128 GB SSD.
- Internet access

3.6 Software Requirements:

- MERN Stack: MongoDB, Express.js, React.js, Node.js
- Socket.IO: Used for real-time chat functionality
- MongoDB: Database management system for data storage
- MUI5: React UI framework for designing user interfaces
- Tailwind CSS: Styling framework for UI elements
- VS Code: For text editor

3.7 Use Case Diagram of the System

Use case symbols

Key components of a use case:

- Actor: An actor is someone who is not a part of the system.
- Use case: A use case is a representation of a system function or operation. It is titled with the function and is depicted as an oval.
- System Boundary: A system is a rectangle-shaped collection of actions that take place when a user interacts with it. This component is optional yet helpful when illustrating complex systems.

Relationship: Relationship is an association between use case and actor

- Symbols of Use case

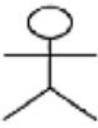
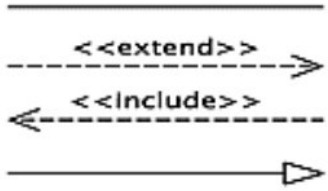
Symbol	Reference Name
	Actor
	Use case
	Relationship

Figure 3.1: Use case Symbols

Multivendor Ecommerce Application for Cloudly Infotech Limited

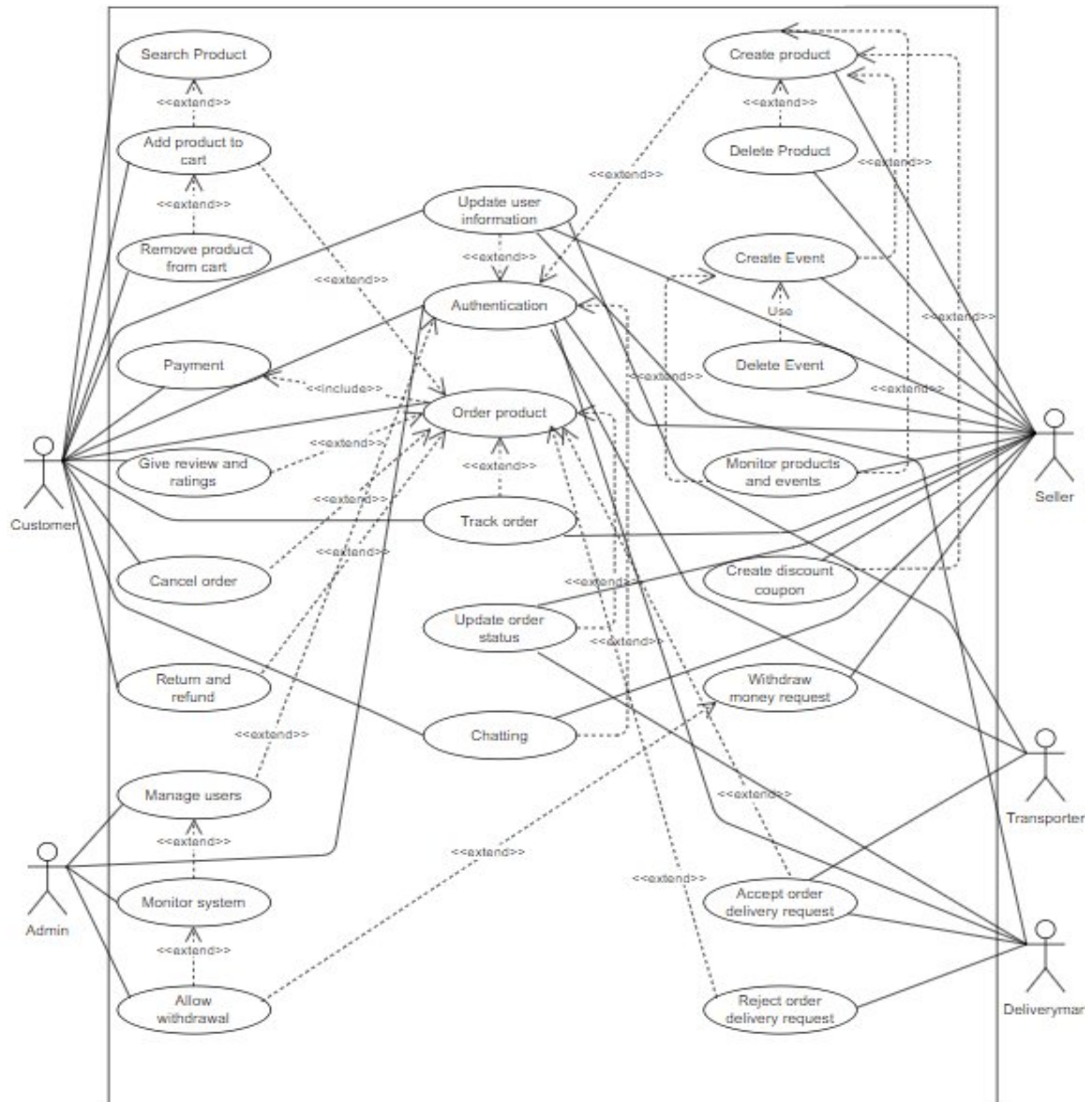


Figure 3.2: Use case Diagram

Chapter 4

System Planning

This project planning chapter provides an overview of the project's functions, along with function point estimation, effort distribution, and a project timetable diagram. The project's goal is to please the client, make it simple for non-technical users to use, and adhere to the following standards:

- Web-based software that can be accessed using any web browser.
- Compatible with Windows, Macs, and Linux; platform independent.

4.1 System Project Planning

Before starting any project, including software projects, it is crucial to engage in project planning activities. Software project planning involves several key activities that help estimate the work, required resources, and time duration, and analyze the feasibility of the project. These activities ensure that the project is well-understood, properly scoped, and has a realistic plan for execution.

The activities of this project's software project planning that have come after are as follows:

- System Project Estimation
- Function Point Estimation
- Based Estimation
- Effort Distribution
- Task Scheduling Project
- Schedule Chart Cost
- Estimation

4.2 System Project Estimation

For projects to be planned and carried out successfully, software project estimating is essential. A software project estimate's accuracy is influenced by a number of variables:

Properly estimated size of the product: Estimating the size of the software product to be built is essential. This involves breaking down the project into smaller components and determining the effort required for each component. Accurate size estimation helps in resource allocation, scheduling, and budgeting.

Alignment with the software team's talents: The project plan should take the involved software team or engineer's abilities and skills into consideration. This involves taking into account their level of productivity, experience, and competence. The accuracy of the estimate is improved by matching the project schedule to the team's capabilities.

Product requirements' stability and the context in which they are developed are key determinants in the success of software engineering efforts. Project delays, additional work, and higher expenses might result from changes in needs or an uncertain environment. Before estimating the project, it is critical to create a reliable baseline for needs.

Software size estimation is indeed a vital aspect of software project estimation. It helps in determining the effort, resources, and schedule required for the project. Inaccurate size estimation can lead to various problems such as scheduling issues, budget overruns, and resource misalignment. To ensure accurate size estimation, it is important to define a bounded software scope, meaning clearly defining the features and functionality that will be included in the software product. By considering these factors and conducting proper software project estimation, project managers can improve the accuracy of their estimates, minimize risks, and ensure the project is executed within the allocated resources and timeframe.

El Table

FTR's	DATA ELEMENTS		
	1-4	5-15	> 15
0-1	Low	Low	Ave
2	Low	Ave	High
3 or more	Ave	High	High

Table 4.1: Complexity Matrix 1

Shared EO and EQ Table

FTR's	DATA ELEMENTS		
	1-5	6-19	> 19
0-1	Low	Low	Ave
2-3	Low	Ave	High
> 3	Ave	High	High

Table 4.2 Complexity Matrix 2

Rating	VALUES		
	EO	EQ	EI
Low	4	3	3
Average	5	4	4
High	7	6	6

Table 4.3 Complexity Matrix 3

RET's	DATA ELEMENTS		
	1-19	20 - 50	> 50
1	Low	Low	Ave
2-5	Low	Ave	High
> 5	Ave	High	High

Rating	Values	
	ILF	EIF
Low	7	5
Average	10	7
High	15	10

Table 4.4 Complexity Matrix 4

Table 4.5 Complexity Matrix 5

4.3 Identifying Complexity

Identifying the complexity of the project is an important step in software project planning. One approach to assessing complexity is by counting function points. Function points are a measure of the functionality provided by a software application. They help estimate the size and effort required for development.

In function point analysis, there are two main categories of functions: Transactional Functions and Data Functions.

1. Transactional Functions:

- External Inputs (EI): These are user-initiated inputs that enter the system. They involve processing data and updating internal files.
- External Outputs (EO): These are the system's responses or outputs to the user. They typically include screens, reports, and files.
- External Queries (EQ): These are inquiries or requests for information from the user, which result in the retrieval of data.

2. Data Functions:

- Internal Logical Files (ILF): These represent internal files or data maintained by the system. They are used for storing and retrieving data within the application.
- External Interface Files (EIF): These are files used by the system but are maintained by external applications or systems. Apart from counting function points, other metrics and factors are considered to analyze the complexity of data functions and transactional functions. Some of these include:
 - File Type Referenced (FTR): It represents the number of different file types referenced by a transaction or a data function.
 - Data Element Types (DET): It represents the number of unique data elements that are referenced or used by a transaction or a data function.
 - Record Element Types (RET): It represents the number of unique record element types within a transaction or a data function.
 - File Element Types (FET): It represents the number of unique file element types referenced by a transaction or a data function.

By considering these metrics and factors, software project planners can assess the complexity of the project and make informed decisions regarding resource allocation, scheduling, and effort estimation.

4.4 Function Point Estimation

Table 4.6: Function, Input and output (Customer)

Customer		
Functionalities	Input	Output
Search Product	Product name	Product details
Add product to cart	Select item add to cart option	Store item in cart
Order Products	Select checkout	Redirect to payment page
Pay bill	Select payment method and give credentials	Save the order in database
Login	Give credentials	Redirect to profile
Register	Give credentials	Redirect to login page
Update profile	Give updated info	Save updated info in database
Give reviews	Write review and select stars for ratings	Save the product review in database
View Orders	Select orders menu	See list of orders been placed
Message with shop	Give message to seller	Seller reply

Table 4.7: Function, Input and output (Shop)

Shop		
Functionalities	Input	Output
Create product	Product info	Save product into database
Update product	Updated product info	Save product into database
Delete product	Select product	Delete product from database
Create event	Product event	Save event into database
Update event	Updated product event	Save event into database
Delete event	Select event	Delete event from database
Update order status	Updated status	Save updated status in database
Login	Give credentials	Redirect to profile
Register	Give credentials	Redirect to login page
Assign order	Select transporter for order	Save shipping info and notify transporter
Message with customer	Reply to customer	Replay from customer
Give withdraw request	Select withdraw account	Save request
Update shop profile	Updated info	Save updated details

Table 4.8: Function, Input and output (Admin)

Admin		
Functionalities	Input	Output
Login	Give credentials	Redirect to profile
Register	Give credentials	Redirect to login page
Manage withdraw requests	Accept or reject	Notify Seller
Monitor Products	Select all products menu	Redirect to all products page
Manage Users	Select all users menu	Redirect to all users page

Table 4.9: Function, Input and output (Transporter)

Transporter		
Functionalities	Input	Output
Login	Give credentials	Redirect to profile
Register	Give credentials	Redirect to login page
View assigned order	Select all orders for menu	Redirect to order details page
Update order status	Update status	Save order status
Functionalities	Input	Output

Table 4.10: Function, Input and output (Deliveryman)

Deliveryman		
Functionalities	Input	Output
Login	Give credentials	Redirect to profile
Register	Give credentials	Redirect to login page
View assigned order	Select all orders for menu	Redirect to order details page
Update order status	Update status	Save order status

Table 4.11: Identifying Complexity (Transactional Function)

Transaction Function	Field/File Involve	FTRs	DETs
Add products (EI)	Fields: product_id, name, description, stock, price, category, images. Files: Products, Shop	2	6
Add Events (EI)	Fields: event_id, name, description, stock, original price, discount price, category, images. Files: Products, Events.	2	7
Update products (EI)	Fields: product_id, name, description, stock, price, category, images. Files: Products, Shop	2	6
Update Events (EI)	Fields: event_id, name, description, stock, original price, discount price, category, images. Files: Products, Events.	2	7
Order Products	Fields: product_id, customer_id, shipping address,	6	7

(EI)	payment_details, shop_id, transporter_id, deliveryman_id Files: Products, Shops, Orders, Transporter, Deliveryman, Customer		
Add Transporter (EI)	Fields: name, address, email, password, phone_number Fields: Transporter	1	6
Add Delivery- man (EI)	Fields: name, address, email, password, phone_number Fields: Transporter	1	6
View Shop (EQ)	Fields: shop_id, user_name File: Shop	1	2
Manage Users (EI)	Fields: user_id, user_email, role, joined_at File: Customer, Shop , Transporter, Deliveryman	4	4
Manage Products (EI)	Fields: id, name, price, stock, ratings, product info File: Products, Shops	2	6
View Orders (EQ)	Fields: id, customer_id, status, product_id, transporter_id, shop_id, deliveryman_id File: Product, Shop, Customer, Orders, Deliveryman, Transporter	6	6
Reviews (EI)	Fields: product_id, ratings, reviews. File: Product	1	3
Manage Users (EI)	Fields: user_id, user_email, role, joined_at File: Customer, Shop , Transporter, Deliveryman	4	4
Manage Products (EI)	Fields: id, name, price, stock, ratings, product info File: Products, Shops	2	6
View Orders (EQ)	Fields: id, customer_id, status, product_id, transporter_id, shop_id, deliveryman_id File: Product, Shop, Customer, Orders, Deliveryman, Transporter	6	6

Table 4.12: Identifying Complexity (Data Function)

Data Function	Fields/File involve	RET's	DET's
Login for system (ILF)	Field: user email, password File: User, Shop, Transporter & Deliveryman	2	4
Earnings (ILF)	Fields: total balance File: Shop	1	1
Total Product (ILF)	Field: product_id, shop_id File: Product, Shop	1	2
Total Orders (ILF)	Fields: product_id, shop_id, customer_id, transporter_id, deliveryman_id, order_id File: Product, Orders, Customer, Deliveryman, Transporter, Shop.	6	6
Approve withdrawal (ILF)	Fields: shop_id, amount, bank info. File: Shop, Withdrawal	2	2

Table 4.13: Unadjusted Function Point Contribution (Transactional Function)

Transaction Function	Field/File Involve	FTRs	DETs
Add products (EI)	Fields: product_id, name, description, stock, price, category, images. Files: Shop, Products	2	6
Add Events (EI)	Fields: event_id, name, description, stock, original price, discount price, category, images. Files: Products, Events.	2	7
Update products (EI)	Fields: product_id, name, description, stock, price, category, images. Files: Products, Shop	2	6
Update Events (EI)	Fields: event_id, name, description, stock, original price, discount price, category, images.	2	7

	Files: Products, Events.		
Order Products (EI)	Fields: product_id, customer_id, shipping address, payment_details, shop_id, transporter_id, deliveryman_id Files: Products, Shops, Orders, Transporter, Deliveryman, Customer	6	7
Add Transporter (EI)	Fields: name, address, email, password, phone_number Fields: Transporter	1	6
Add Delivery-man (EI)	Fields: name, address, email, password, phone_number Fields: Transporter	1	6
View Shop (EQ)	Fields: shop_id, user_name File: Shop	1	2
Manage Users (EI)	Fields: user_id, user_email, role, joined_at File: Customer, Shop , Transporter, Deliveryman	4	4
Manage Products (EI)	Fields: id, name, price, stock, ratings, product info File: Products, Shops	2	6
View Orders (EQ)	Fields: id, customer_id, status, product_id, transporter_id, shop_id, deliveryman_id File: Product, Shop, Customer, Orders, Deliveryman, Transporter	6	6

Table 4.14: Unadjusted Function Point Contribution (Data Function)

Data Function	Fields/File involve	RET's	DET's	Complexity	UFP
Login for system (ILF)	Field: user email, password File: User, Shop, Transporter & Deliveryman	2	4	Low	7
Earnings (ILF)	Fields: total balance File: Shop	1	1	Low	7
Total Product (ILF)	Field: product_id, shop_id File: Product, Shop	1	2	Low	7
Total Orders (ILF)	Fields: product_id, shop_id, customer_id, transporter_id, deliveryman_id, order_id File: Product, Orders, Customer, Deliveryman, Transporter, Shop.	6	6	Avg	10
Approve withdrawal (ILF)	Fields: shop_id, amount, bank info. File: Shop, Withdrawal	2	2	Low	7

Table 4. 15 Performance and Environmental impact

General System Characteristic	TDI
1. Data Communication	2
2. Distributed Data Processing	1
3. Heavily Used Configuration	2
4. Transaction rate	3
5. On-Line data entry	3
6. End-user efficiency	3
7. On-Line update	0
8. Complex processing	2
9. Performance	3
10. Re-usability	2
11. Installation ease	3
12. Operational ease	3
13. Multiple sites	0
14. Facilitate change	3
Total Degree of Influence	30

4.5 Calculations:

To calculate the function points and estimate the effort required for the project, you have provided the

following information:

Value Adjustment Factor (VAF) calculation:

$$\text{VAF} = (0.65 + (0.01 \times \text{TDI}))$$

where TDI is the Technical Difficulty Index, given as 30.

$$\text{VAF} = (0.65 + (0.01 \times 30)) = 0.95$$

Counting Adjusted Function Point:

$$\text{UFP} = \text{UFP (Data function)} + \text{UFP (Transaction Function)}$$

$$\text{UFP (Data function)} = 38$$

$$\text{UFP (Transaction Function)} = 56$$

$$\text{UFP} = 56 + 38 = 94$$

Adjusted Function Point Count:

$$\text{Adjusted Function Point Count} = \text{UFP} \times \text{VAF}$$

$$\text{Adjusted Function Point Count} = 94 \times 0.95 = 93.04$$

Productivity for JavaScript:

$$\text{Productivity for JavaScript} = 15.5 \text{ (given)}$$

Effort for JavaScript:

$$\text{Effort for JavaScript} = \text{Adjusted Function Point Count} \times \text{Productivity for JavaScript}$$

$$\text{Effort for JavaScript} = 93.4 \times 15.5 = 1447.7 \text{ person hours}$$

Considering 8 hours of work per day:

$$\text{Effort for JavaScript} = 1447.7 \text{ person hours} / 8 \text{ hours} = 181 \text{ Person Days}$$

Converting to person months:

Effort for JavaScript = $181 \text{ Person Days} / 24 = 7.5 \text{ Person Months}$

Converting to a team of 4 persons:

Effort for JavaScript = $7.5 \text{ Person Months} / 4 = \text{Months for } 1.29 \text{ persons}$

Therefore, the estimated effort for the project is approximately Person Days or Person Months.

Considering a team of 4 persons, the estimated duration is around 2 months. And for a single person almost 7.5 months.

4.6 Effort Distribution

In the software project estimation for this project, the effort distribution has been allocated as follows:

Coding/Implementation: 40% Analysis and Design: 23% Software Testing and Support: 37%.

These percentages indicate the proportion of effort allocated to each phase of the software development process.

Based on the total estimated effort for the project, you can distribute the effort according to these percentages to determine the number of work units allocated to each phase. For example, if the total estimated effort is 100 units:

- Coding/Implementation: 40 units
- Analysis and Design: 23 units
- Software Testing and Support: 37 units

By distributing the effort in this manner, you can ensure that the necessary resources and time are allocated to each phase of the software development process in accordance with the project plan.

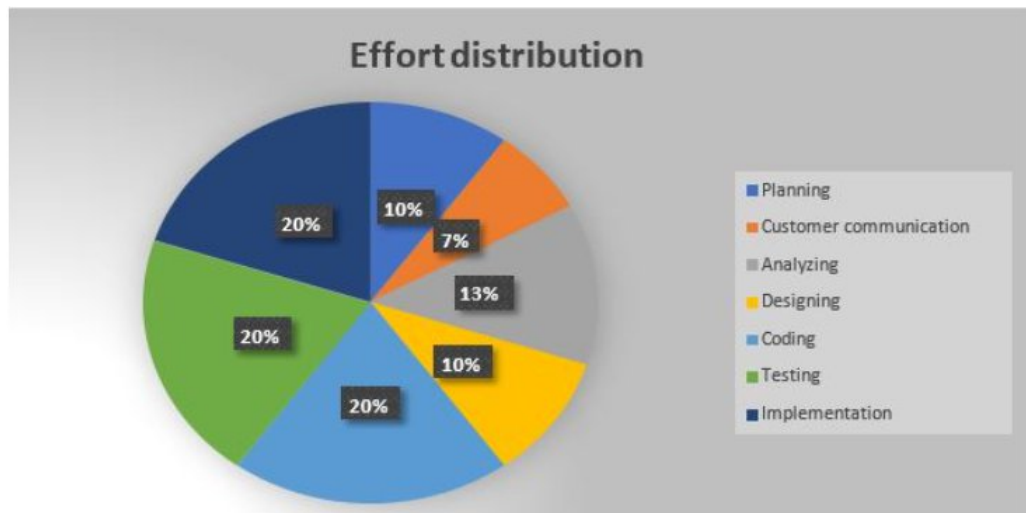


Figure 4.1: Effort Distribution Estimation

4.7 Time schedule

A variety of sets of jobs are combined to create a total system. These responsibilities must be completed in a timely manner and in order. The system developer uses the project schedule as a guideline. The project's schedule is shown in the table below:

Task	Time	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15	W16
Phase One	Project Planning																
	Requirements Specification																
	Design Specifications																
	Initial version Development																
	Project Test Plan																
Phase Two	Final Requirements Specification																
	Software Development																
	Phase Review																
Phase Three	Final Design Specifications																
	Final version Development																
	Validation Testing																
	Phase Review and Approval																

Figure 4.2: Time schedule chart

4.8 Cost Estimation

Personnel cost

To calculate the personnel cost for the project, we need to consider the number of working days and working hours involved. In a year, we have 365 days, which includes 24 government holidays and 52 weekly holidays. Subtracting these from the total gives us 289 working days specifically for the project. Dividing this by 12 months, we get an average of approximately 24.083 working days per month.

Considering that the organization works for 8 hours per day, the total working hours per month can be calculated by multiplying the average working days by the daily working hours. In this case, it would be 24.083 multiplied by 8, resulting in approximately 192.664 working hours per month.

For the duration of the project, which is estimated to be around 2 months, the total working hours can be obtained by multiplying the monthly working hours by the number of months. In this case, it would be 192.664 multiplied by 8 and then by 2, giving us an approximate total of 385.328 working hours.

Table 4.16: Personal cost

Position	Salary per month	Salary per hour
System analyst	35000	181.66
Designer	25000	129.75
Coder	20000	103.80
Tester	20000	103.80

Total salary of Analyst in 2 months $160 \times 181.66 = 29065.6\text{TK}$

Total salary of Designer in 2 months $220 \times 129.75 = 30745\text{TK}$

Total salary of coder in 2 months $300 \times 103.80 = 31140\text{TK}$

Total salary of tester in 2 months $86 \times 103.80 = 8926.8\text{TK}$

Total Salary = 99877.4 TK

Table 4.17: Depreciation Calculation

Number of year	Depreciation Calculation
Year1	$5/15 = 33.33\%$
Year 2	$4/15 = 26.67\%$
Year 3	$3/15 = 20\%$
Year 4	$2/15 = 13.33\%$
Year 5	$1/15 = 6.67\%$

Hardware cost

- ☞ Cost of laptop 35000 TK
- ☞ Lifetime of laptop 5 years
- ☞ The percentage of year is $2/15 = 13.33\% = 0.1333$
- ☞ The depreciation cost of laptop is $= 35,000 * 0.1333 = 4665.5$
- ☞ Printer cost 5000 TK
- ☞ Life time of printer 5 years
- ☞ The percentage of year is $5/15 = 33.33\% = 0.3333$
- ☞ The depreciation cost of printer is $= 5,000 * 0.3333 = 1666.5$
- ☞ Total hardware cost $4665.5 + 1666.5 = 6332$

Software cost

Table 4.18: Software cost

Name of software	cost
Windows 10 pro	400 TK
Microsoft office	100 TK
MongoDB	Free
VS code	Free
Total	500TK

Other cost

Table 4.19: Other cost

Name	Cost
Transport	2000
House rent	12000
Internet	2000
Others	1500
Total	17500 TK

Account table

Table 4.20: Account table

Particulars	Cost
Personal cost	99877.4
Hardware cost	6332
Software cost	500
Other costs	17500
	124209.4 TK

Chapter 5

Risk Management

5.1 Risk Management

Risk management is a crucial aspect of any software project as it involves identifying and addressing potential risks that may arise. Failing to properly analyze and estimate project risks can lead to numerous problems. Therefore, it is important to implement effective risk management strategies.

The process of risk management consists of several stages aimed at understanding and managing uncertainties. These stages help the software team in identifying, assessing, and mitigating risks.

The following phases are typically followed to establish a risk management model:

Identification: This entails gathering data in order to identify potential dangers. Information regarding potential risks and hazards can be gathered using a variety of automated and human ways.

Classification: Risk classification requires the creation of a structured model to classify risks and the alignment of observable risk characteristics and occurrences with the model. It is possible to assess and categorize hazards using both quantitative and qualitative methods.

Assessment: Risk assessment focuses on identifying pertinent risk scenarios or events that may result in loss or damage. It entails calculating the likelihood that these occurrences will occur. A thorough risk analysis should provide appropriate recommendations and be transparent, logical, and consistent.

Analysis: The possible effects of risk patterns or scenarios are determined by risk analysis, including the size of prospective losses and the recovery costs included. It analyzes weaknesses and takes into account the organization's risk tolerance, which results in the creation of mitigation measures.

Implementation: In order to manage and address identified risks, risk management implementation requires developing policies, processes, and methods. It should establish a balance between the expense of preventing damage or loss and the direct and indirect costs

associated with maintaining the value of assets. Considerations for web-based systems should include the environment's hardware and

software, administrative practices, load balancing, backup media selection, server vulnerabilities, and defense against various attacks.

By following a comprehensive risk management approach, software projects can effectively address potential risks and minimize their impact on the project's success.

5.2 Categories of Risk

It's crucial to take into account a variety of risk categories in any software project. In this software project, the following risk categories have been taken into consideration:

1. Project Risks: The project plan is in danger because of these dangers. If these come to pass, the project's schedule will probably be delayed, and costs could go up. Assessing project risks entails detecting potential problems with the software project's budget, timeline, Seller, resources, customers, and requirements.

2. Technical Risks: These dangers put the caliber and timeliness of the program under development in jeopardy. It can become difficult or even impossible to implement the program if a technical risk materializes. Finding possible issues with design, implementation, interfaces, verification, and maintenance are considered technical risks. Technical hazards are also influenced by elements including specification ambiguity, technical uncertainty, and technological obsolescence.

3. Business Risks: These dangers put the program that is being developed at danger of failing. Business risks can take many different shapes, such as market risks (creating a system with insufficient demand), strategic risks (creating a system that no longer aligns with the company's overall business strategy), management risks (losing senior management's support due to shifts in focus or personnel), and budget risks (rising costs or a reduction in Seller commitment).

By recognizing and addressing risks in these categories, the software project can proactively manage uncertainties and make informed decisions to mitigate potential negative impacts.

5.3 The RMMM Plan

The RMMM (Risk Mitigation, Monitoring, and Management) Plan is an essential component of risk management in a software project. It outlines the strategies and actions to address risks throughout the project lifecycle. The following components are included in the RMMM Plan:

- 1. Risk Mitigation:** This involves proactive planning to avoid or minimize the occurrence and impact of identified risks. Mitigation strategies may include preventive measures, contingency plans, alternative approaches, and risk reduction techniques. The goal is to reduce the probability and severity of risks.
- 2. Risk Monitoring:** Continuous assessment of predicted risks to determine if they occur and to ensure that preventive measures are effectively implemented. Monitoring also involves collecting relevant information for future risk analysis and identifying the relationship between risks and their associated problems. By monitoring risks, the project team can take timely actions to address emerging issues.
- 3. Risk Management:** This refers to the actions taken when mitigation measures have failed, and a risk becomes an actual problem. Risk management involves implementing predefined response plans, activating contingency measures, and mobilizing resources to address the risk and minimize its impact on the project. The focus is on resolving the issue and minimizing any negative consequences.

Type of Impact: Impact assessment is a crucial aspect of risk management. The RMMM Plan categorizes the impact of risks into different levels based on their severity. The impact levels can be categorized as follows:

- Catastrophic (1): Risks that have severe consequences, leading to significant disruptions or failures in project objectives.
- Marginal (2): Risks that have minimal impact on the project, causing minor setbacks or delays but can be managed within the project's existing capabilities.
- Tolerable (3): Risks that have a moderate impact on the project, resulting in noticeable challenges and requiring additional resources or adjustments to overcome.
- Critical (4): Risks that have a significant impact on project success and require immediate attention and decisive actions to prevent major failures.

By incorporating risk mitigation, monitoring, and management strategies, as well as considering the impact levels of risks, the RMMM Plan helps ensure effective risk management throughout the software project lifecycle.

Type of Probability: very low (<10%), low (10–25%), moderate (25–50%), high (50–75%), very high (>75%).

Project Risks: In my system, the below mentioned projects risks I needed to manage.

Table 5.1: Project Risk 1

Project risk 1	
Name	Change in requirement
Probability	Low
Impact	Marginal
Description	Customer may change their requirement
Mitigation and Monitoring	Due to time constraints or operational requirements, the corporation redefines requirements. Regular meetings will be held with the company. This guarantees that the issue our product is meant to solve.
Management	Emergency meeting between both parties to identify new project requirements and goals.
Status	Not Occur

Table 5.2: Project Risk 2

Project risk 2	
Name	Poor Quality of documentation
Probability	Catastrophic
Impact	Documentation quality may poor
Description	Customer may change their requirement
Mitigation and Monitoring	A routine meeting will be held to discuss possible documentation topics. Each meeting will also include a Monitor who will track the documentation progress.
Management	A meeting will be scheduled to talk about improving the quality of the documentation. New items will be assigned to responsible individuals.
Status	Monitoring it

Table 5.3: Technical Risk 1

Project risk 3	
Name	Computer crash
Probability	Catastrophic
Impact	Computer may crash due to several reason
Description	We need to properly monitor computers. Use these with caution.
Mitigation and Monitoring	If the computer has been crashed then we will restore backup.
Management	Managed it.
Status	Monitoring it

Chapter 6

Analysis & Modeling

Analysis modeling is a vital stage in software development that represents the specifications for data, function, and behavior using a combination of text and diagrammatic representations. Its goal is to make the system's requirements plain and understood while facilitating the review process for accuracy, completeness, and consistency. Resources for traditional and object-oriented analytic techniques, as well as resources for the Unified Modeling Language (UML), are introduced in this section.

6.1 Software Analysis Pattern:

The objectives of analysis patterns are as follows:

1. Domain Analysis: Analyzing the specific domain in which the software will be used to gain a deeper understanding of its requirements and constraints.
2. Describing Client Requirements: Clearly describing the requirements and expectations of the client or end-users.
3. Establishing a Basis for Software Design: Providing a foundation for creating the software design by identifying key elements and relationships within the system.
4. Specifying a set of criteria that can be verified and validated after the program is constructed is known as “defining valid table requirements.

6.2 Activity Diagram:

Graphical representations of processes comprising sequential activities and events within a system are called “activity diagrams.” They provide the visual depiction of options, iterations, and concurrency.

Both organizational and computational processes can employ activity graphs. Activity diagrams are used in the context of the Unified Modeling Language (UML) to represent the overall control flow inside a system.

Software developers may better comprehend system requirements and successfully create a solution that satisfies the client's expectations by using analytical modeling approaches and tools like activity diagrams.

Activity diagram of Authentication

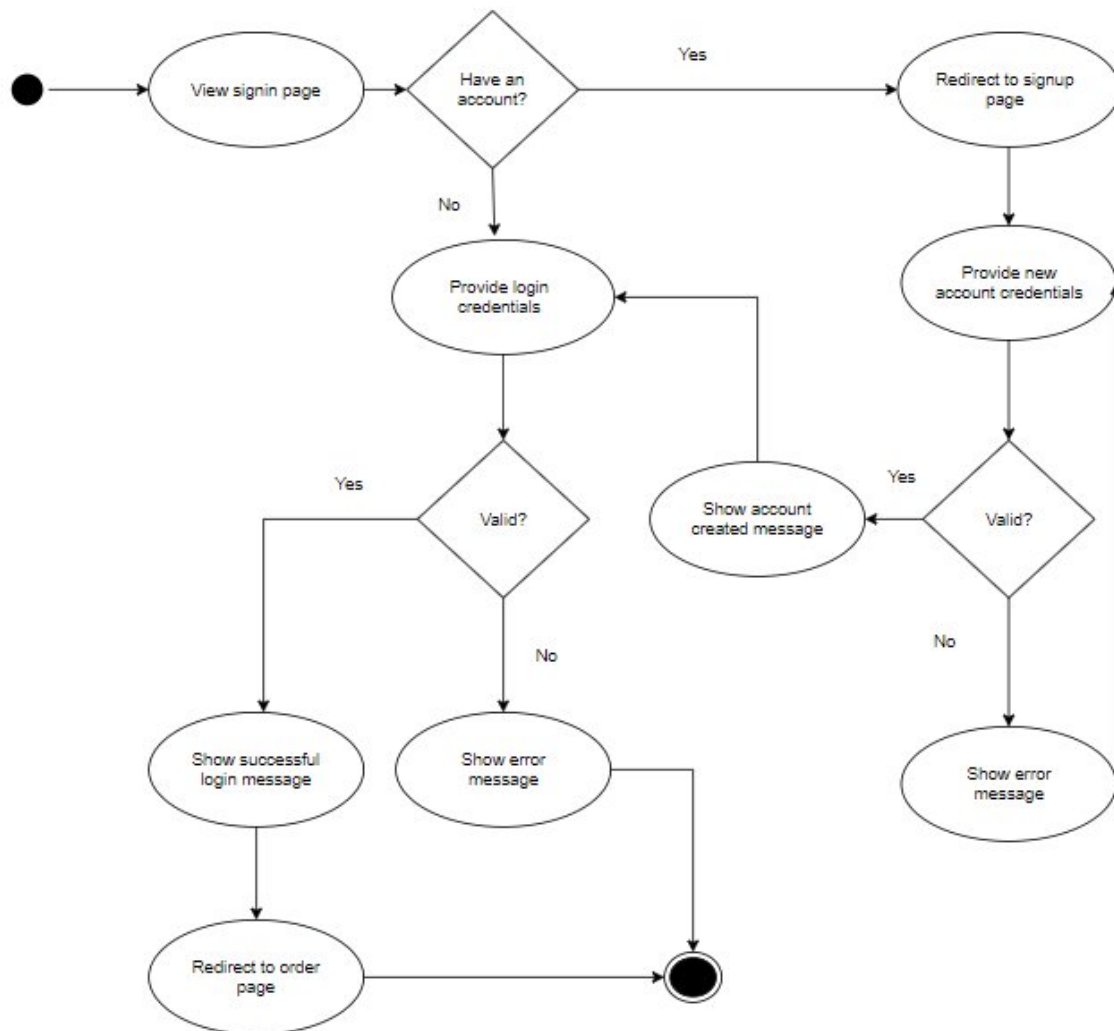


Figure 6.1 Activity diagram for Authentication

Activity diagram of View product

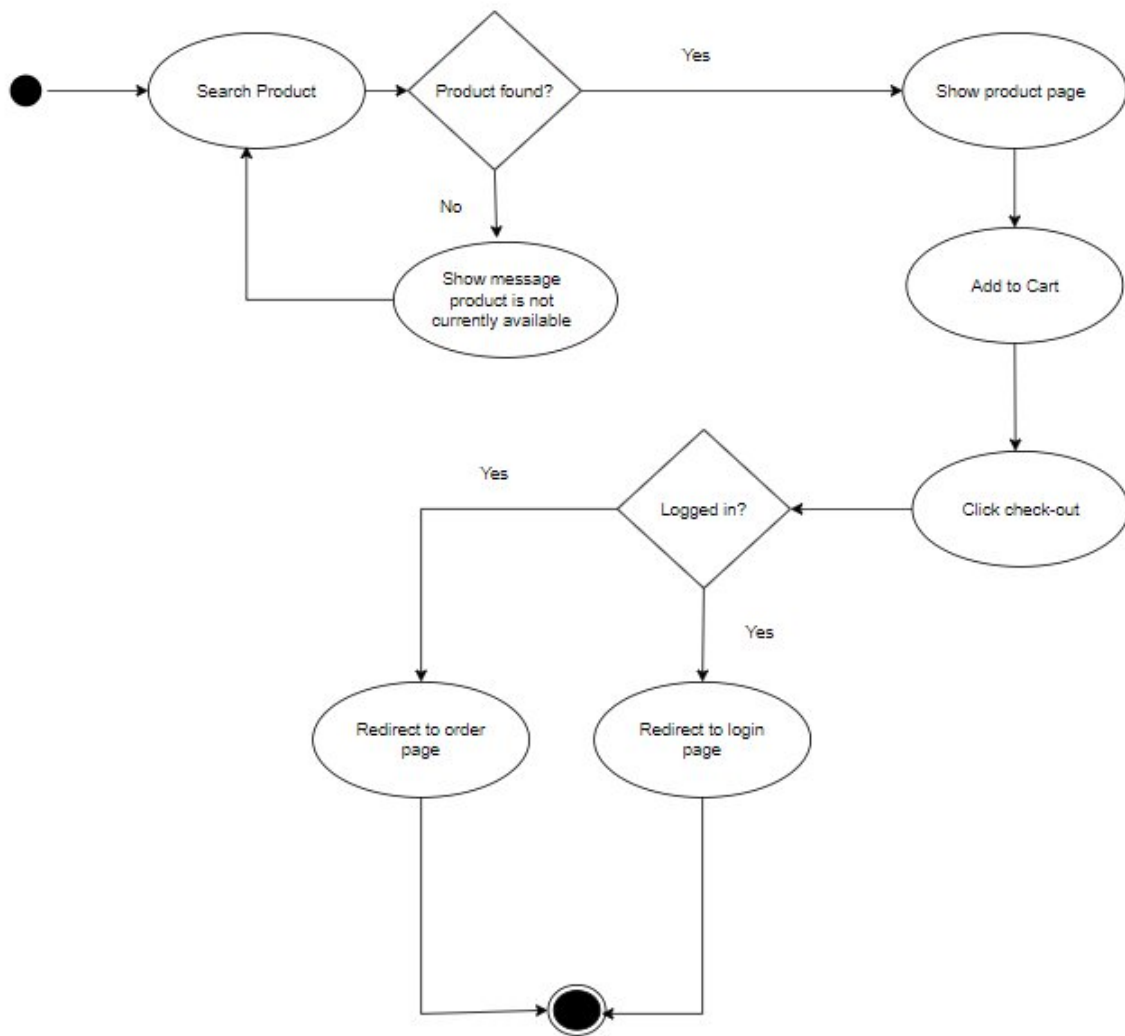


Figure 6.2 Activity diagram of view product

Activity diagram of Order product

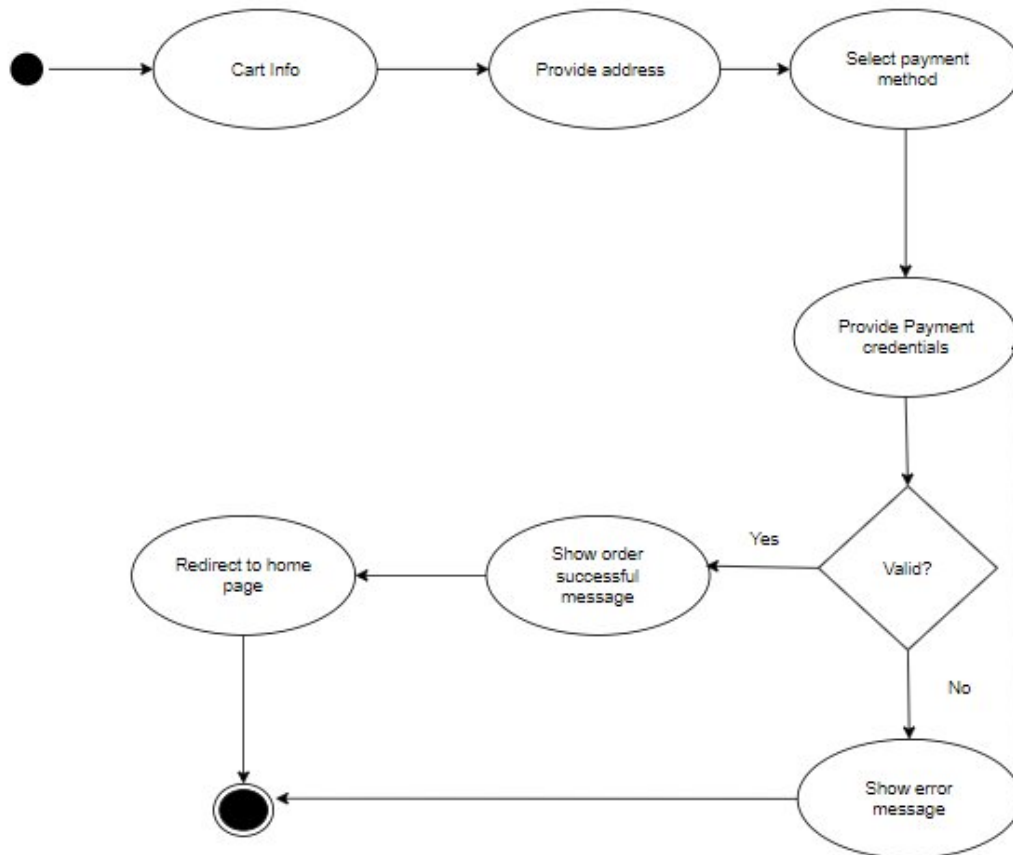


Figure 6.3 Activity diagram of order product

Activity diagram of Shop

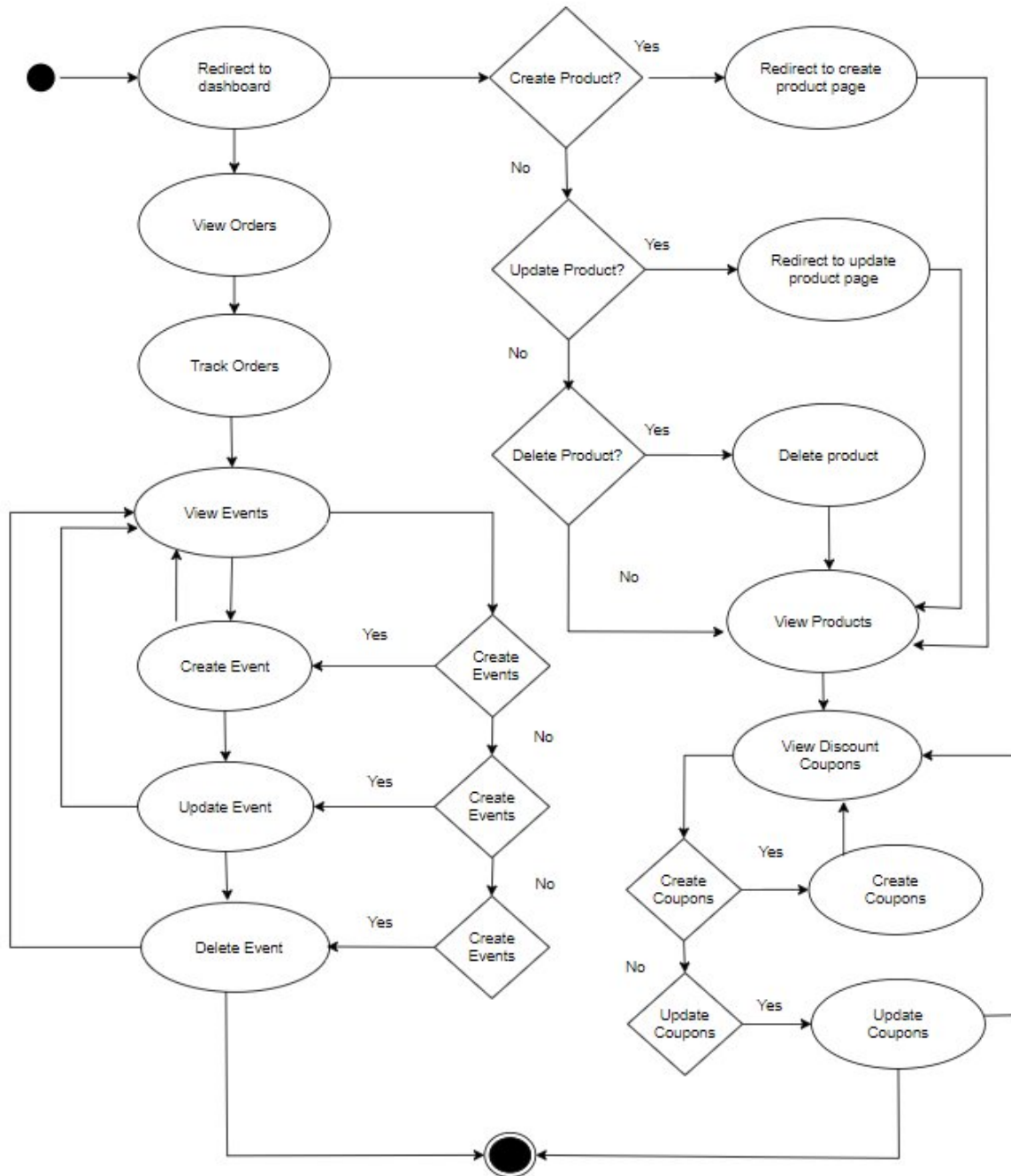


Figure 6.4 Activity diagram for shop management

Activity diagram of Transport and Delivery of product

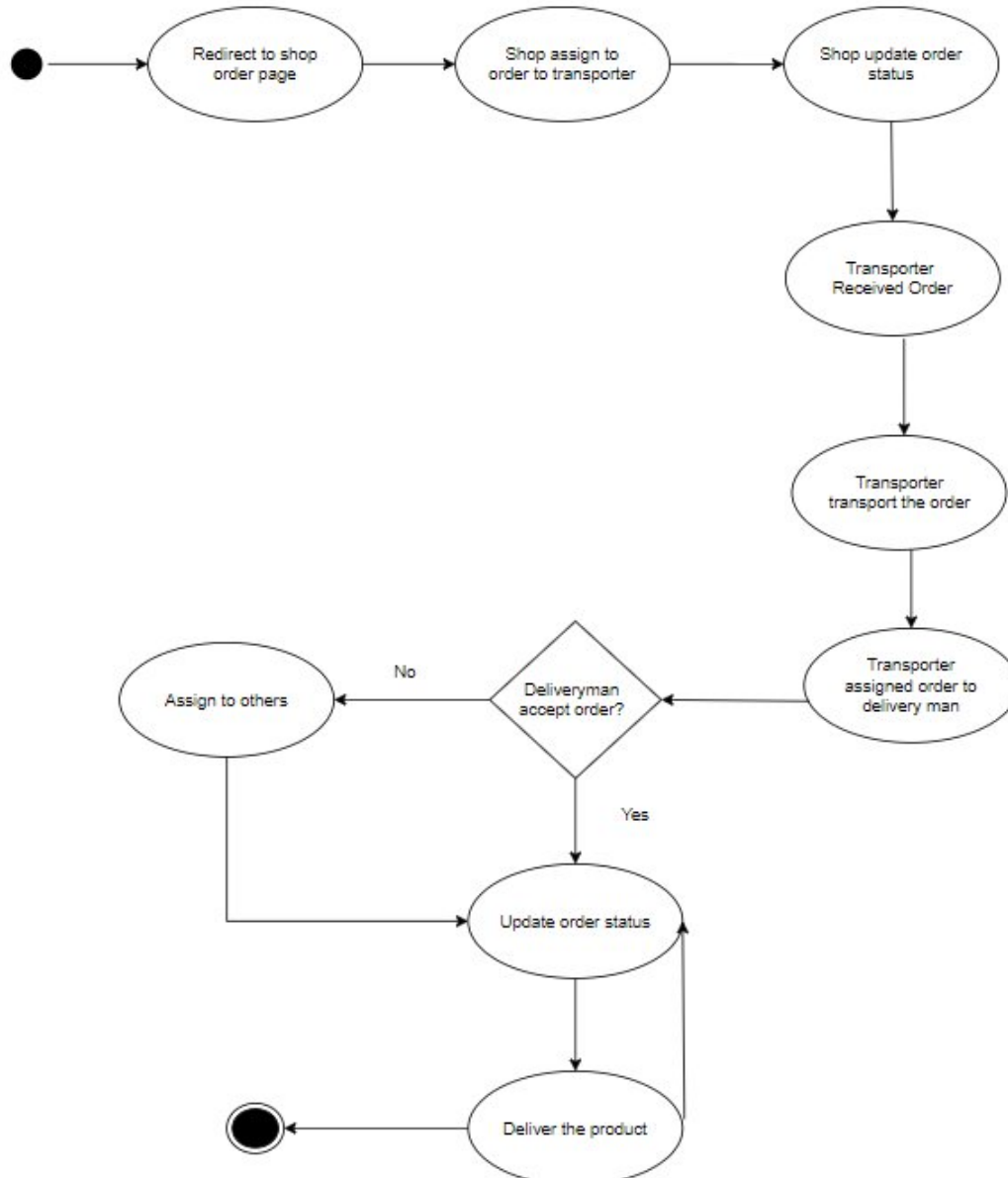


Figure 6.5 Activity diagram for transport and delivery of product

Activity diagram of Refund

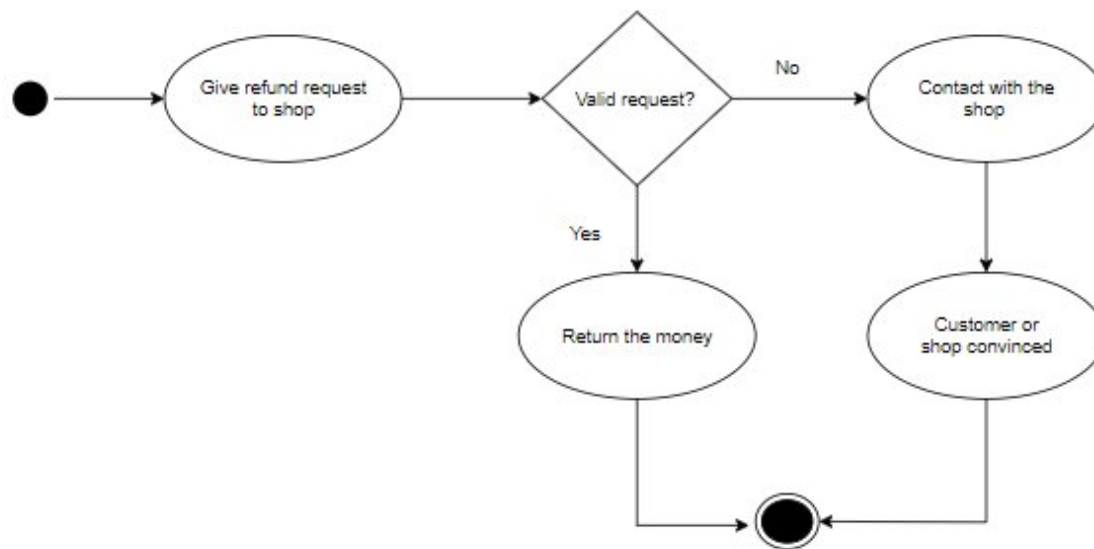


Figure 6.6 Activity diagram for order refund

Activity diagram of Conversation

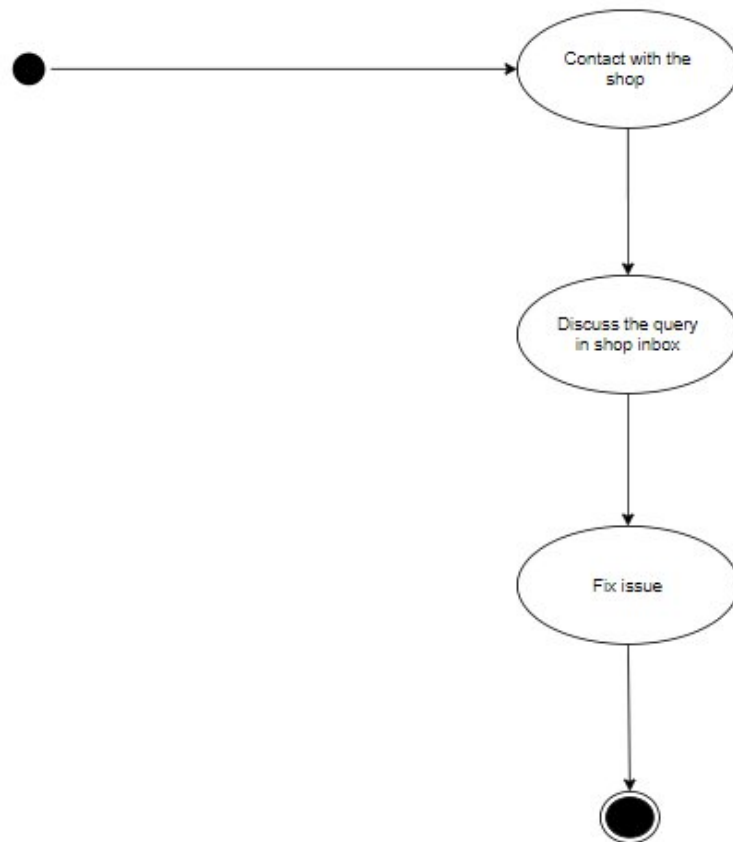


Figure 6.7 Activity diagram for conversation

Activity diagram of Withdraw Money

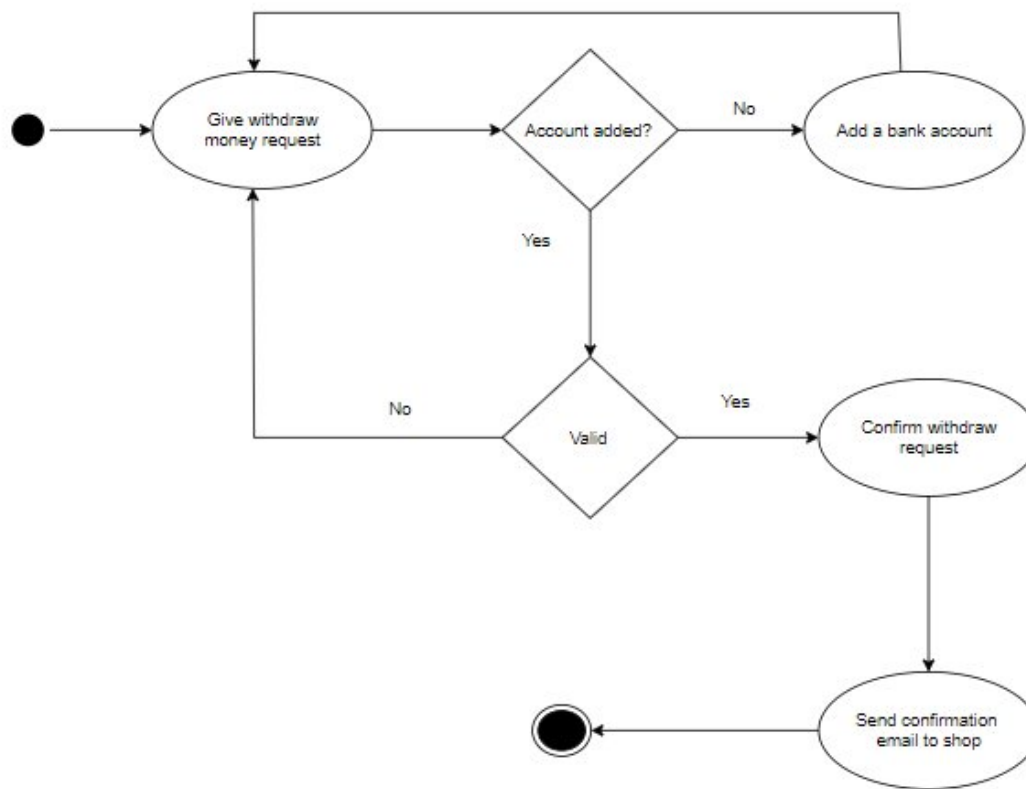


Figure 6.8 Activity diagram for withdraw money

Activity diagram of System Monitoring

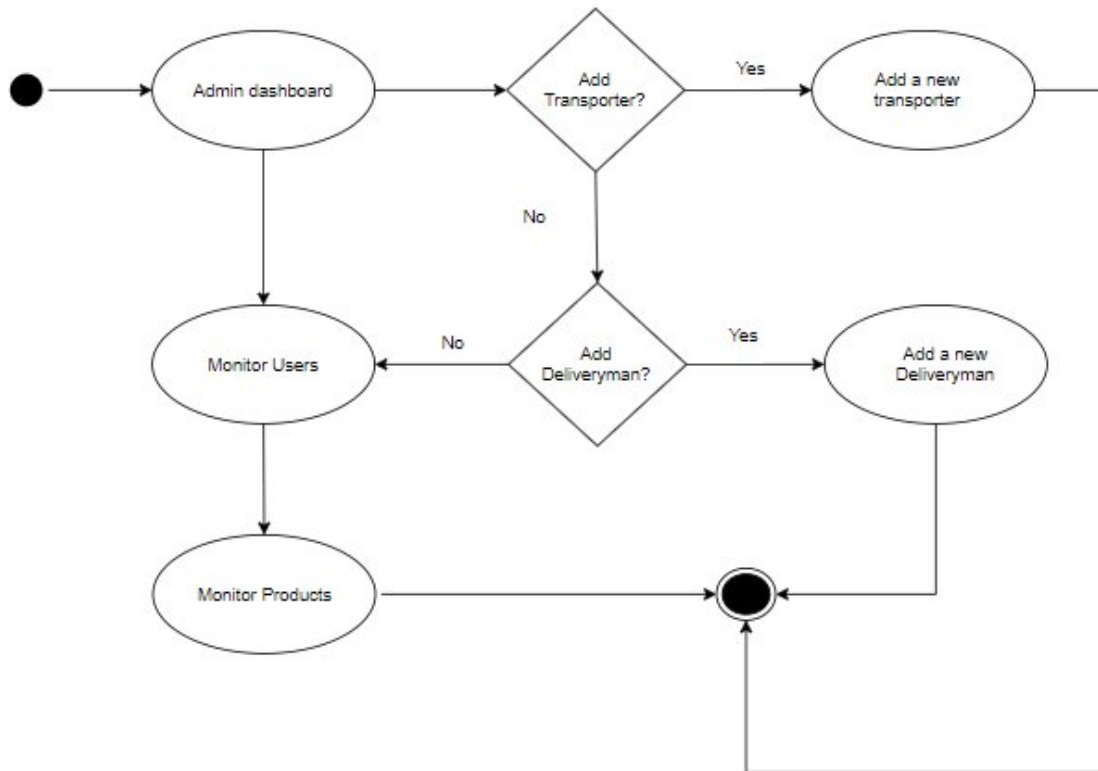


Figure 6.9 Activity diagram for system monitoring

6.3 Swim Lane Diagram

A Swim Lane diagram is a visual representation that illustrates the flow of activities in a process while clearly indicating the responsible individuals, work groups, or departments involved. It gets its name from the lanes resembling separate swimming lanes in a pool, with each lane representing a specific participant or group.

The Swim Lane diagram is used to bring clarity and accountability to a process by assigning process steps to the respective lanes. This allows for a clear understanding of who is responsible for each task or activity. By visually mapping out the process flow within the Swim Lanes, it becomes easier to identify connections, communication channels, and handoffs between different participants.

One of the benefits of using a Swim Lane diagram is that it can help identify areas of waste, redundancy, and inefficiency within a process. By analyzing the interactions between lanes and the sequence of activities, bottlenecks and areas for improvement can be identified more easily.

Overall, Swim Lane diagrams provide a visual representation of a process, enabling stakeholders to better understand the flow of activities, roles, and responsibilities involved. They can be an effective tool for process analysis, optimization, and communication within an organization.

Swim lane diagram of Authentication

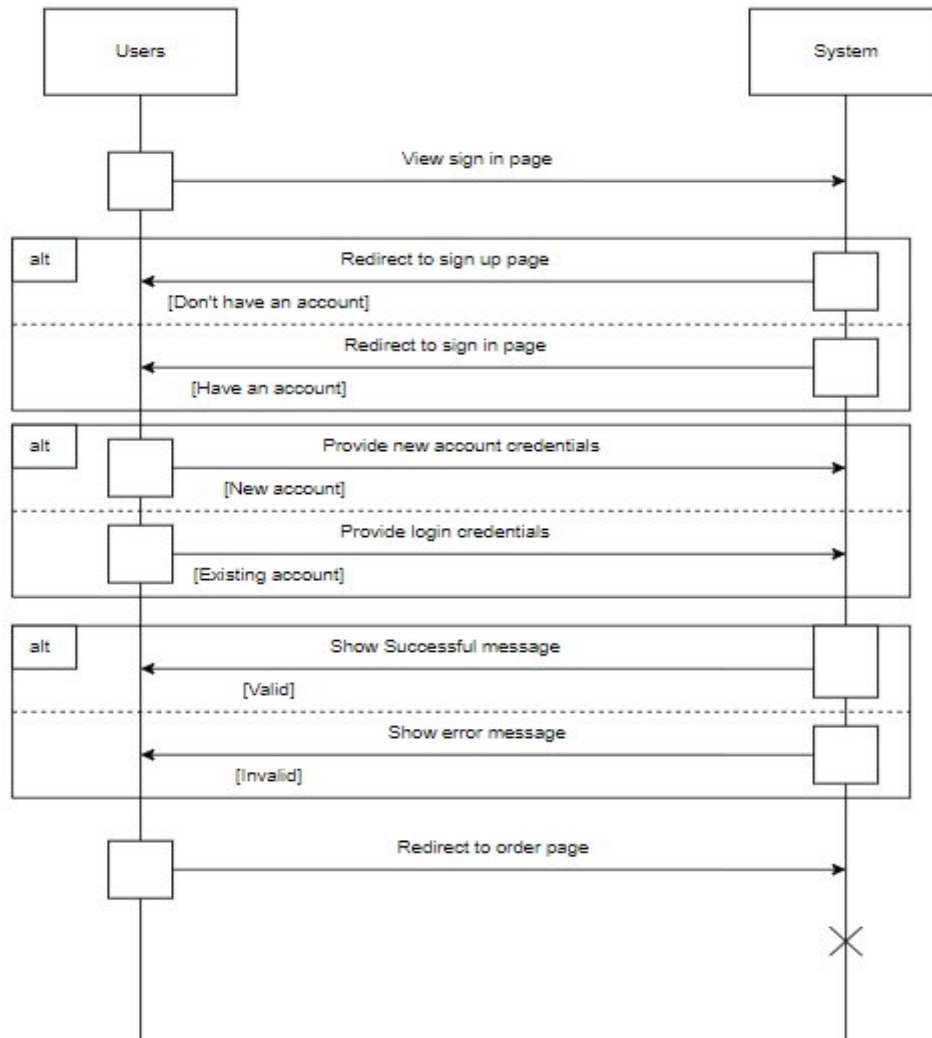


Figure 6.10 Swim Lane diagram for Authentication

Swim lane diagram of View product

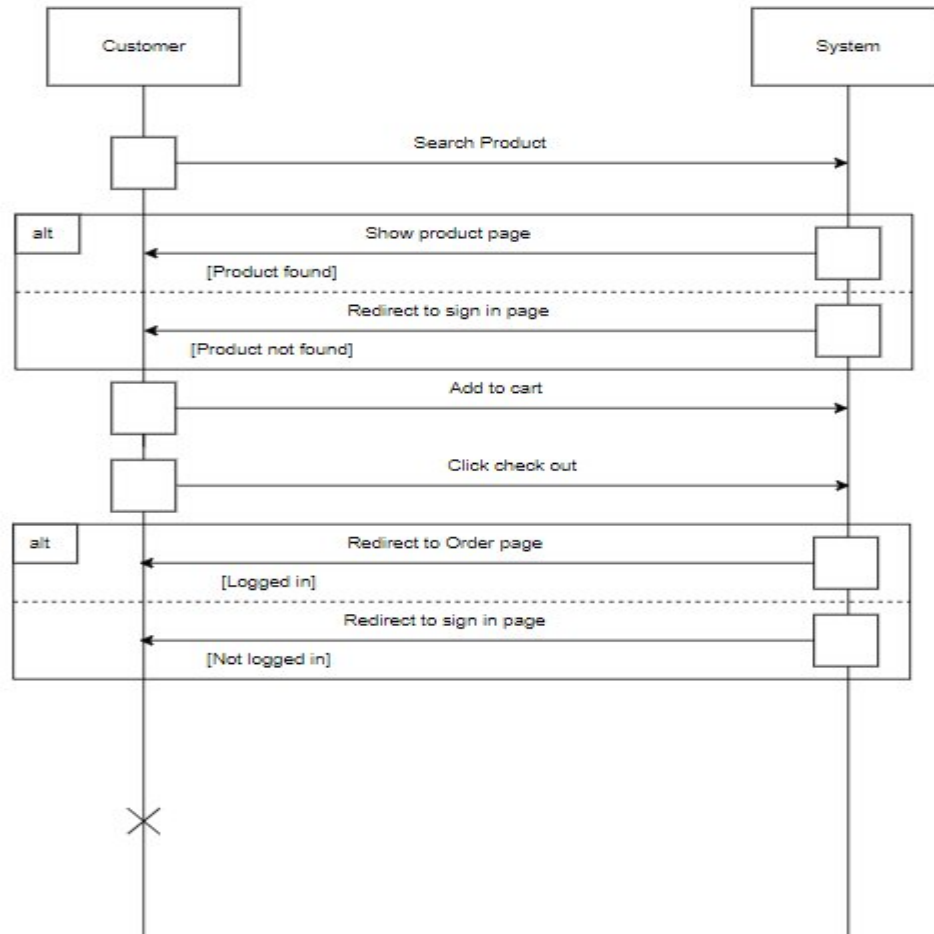


Figure 6.11 Swim Lane diagram for view product

Swim lane diagram of Order product

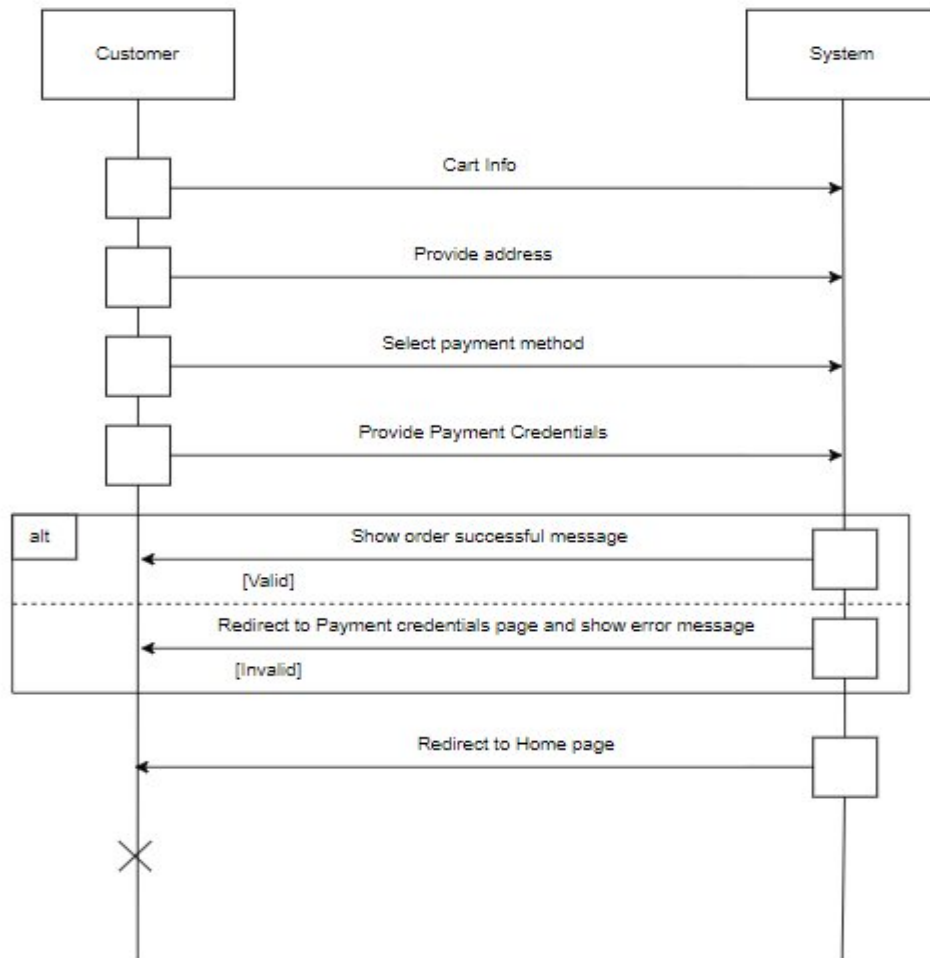


Figure 6.12 Swim Lane diagram for order product

Swim lane diagram of Shop

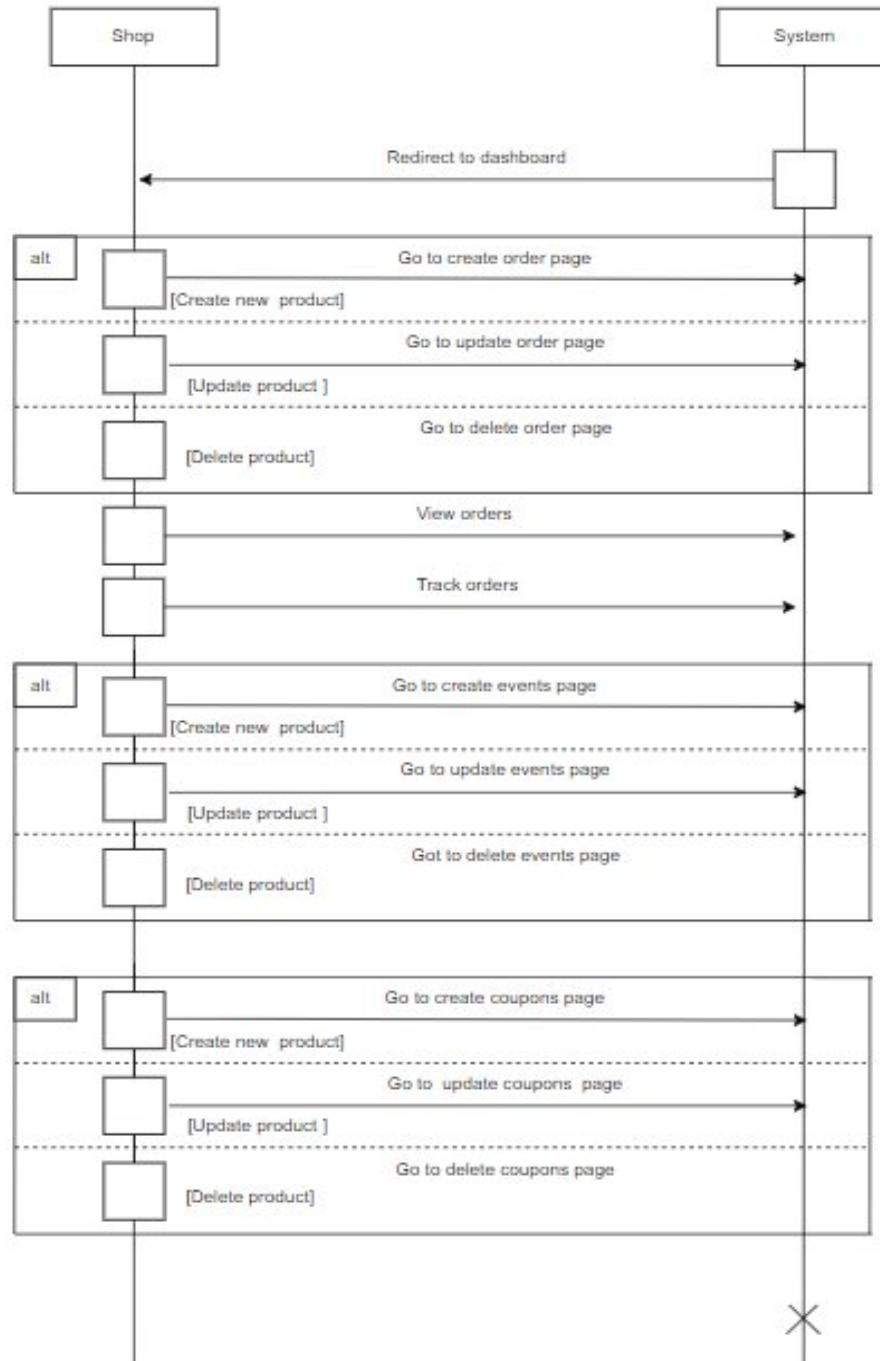


Figure 6.13 Swim Lane diagram for shop management

Swim lane diagram of Transport and Delivery product

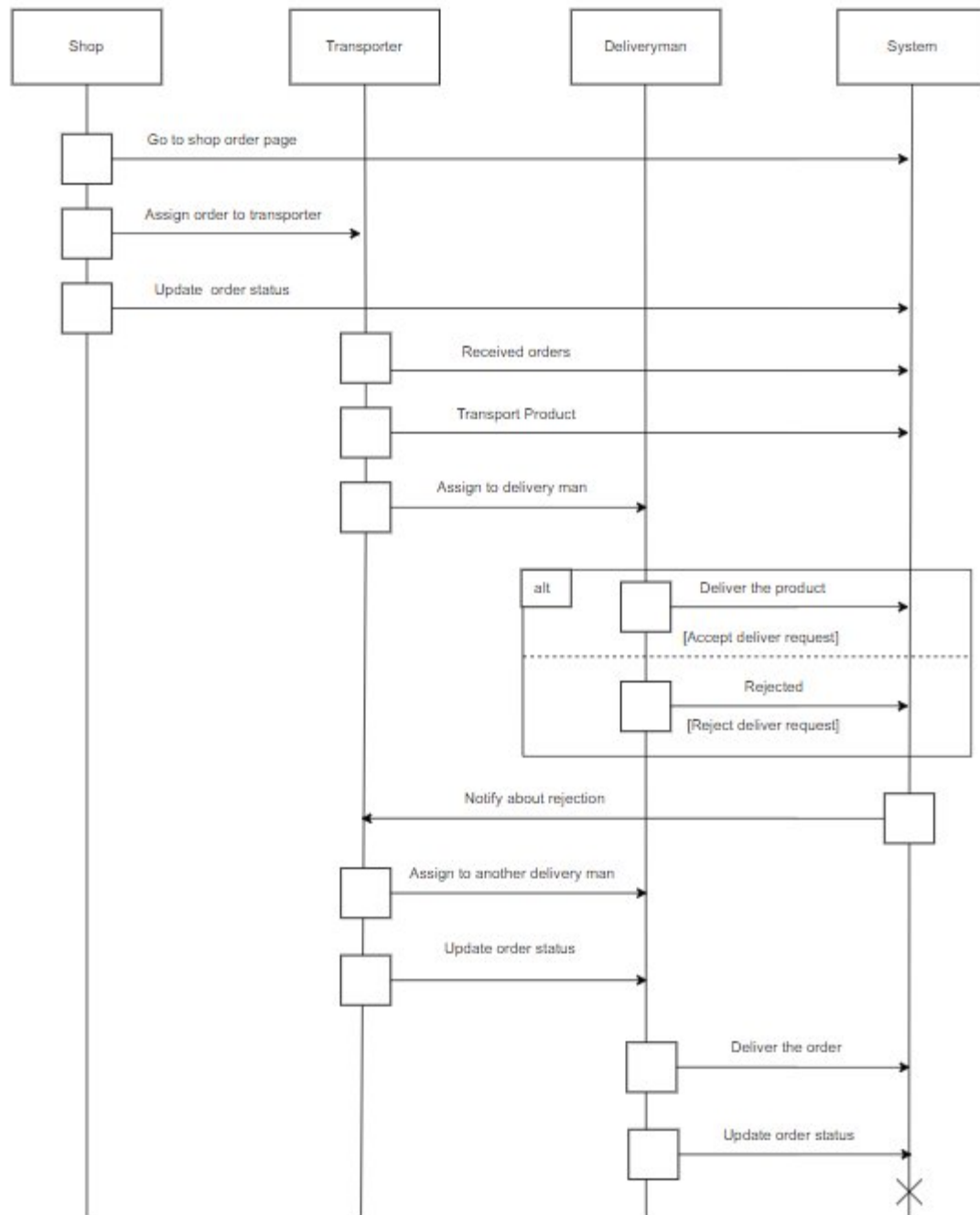


Figure 6.14 Swim Lane diagram for transport and delivery of product

Swim lane diagram of Refund product

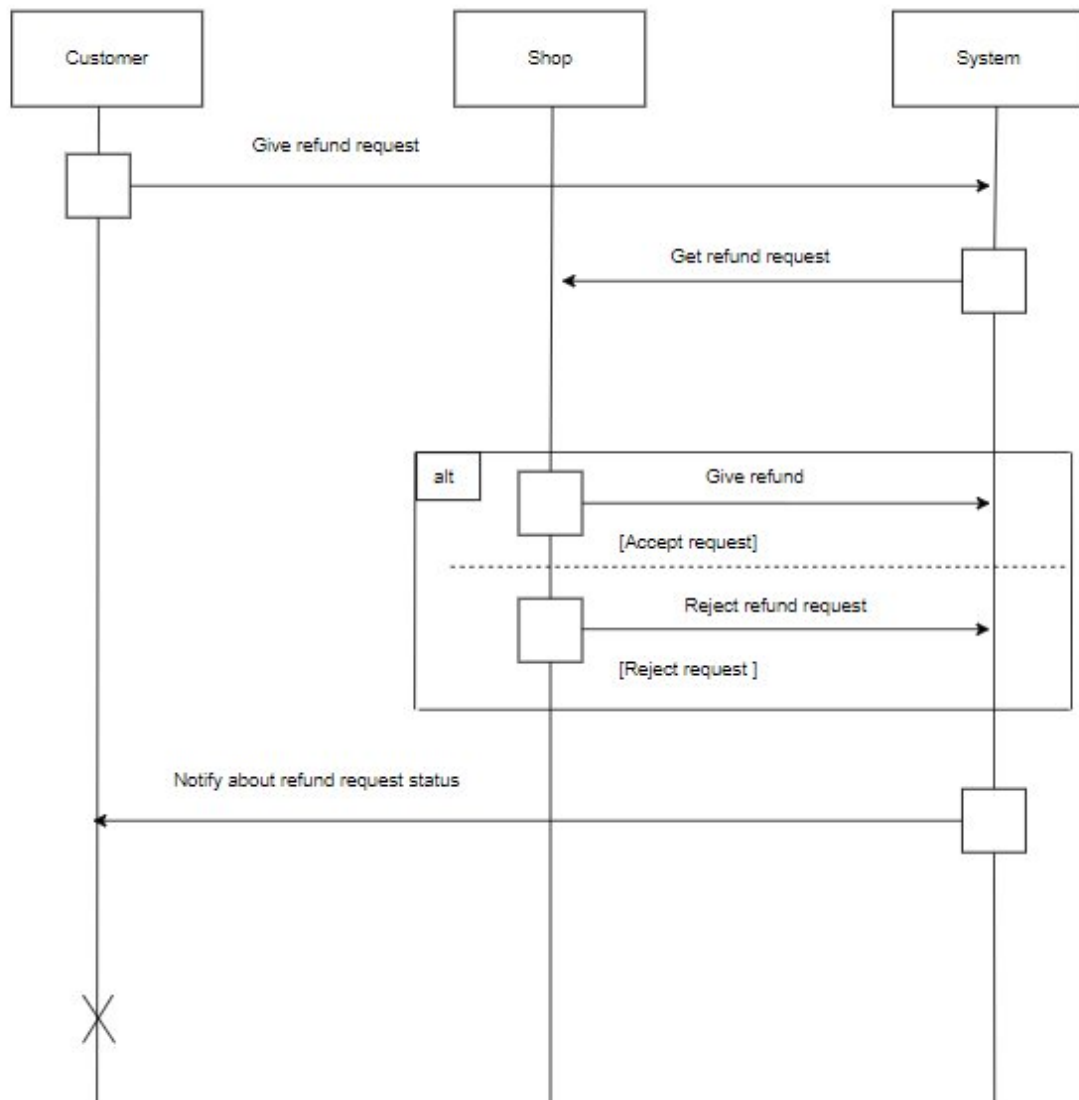


Figure 6.15 Swim Lane diagram for refund and return of product

Swim lane diagram of Conversation

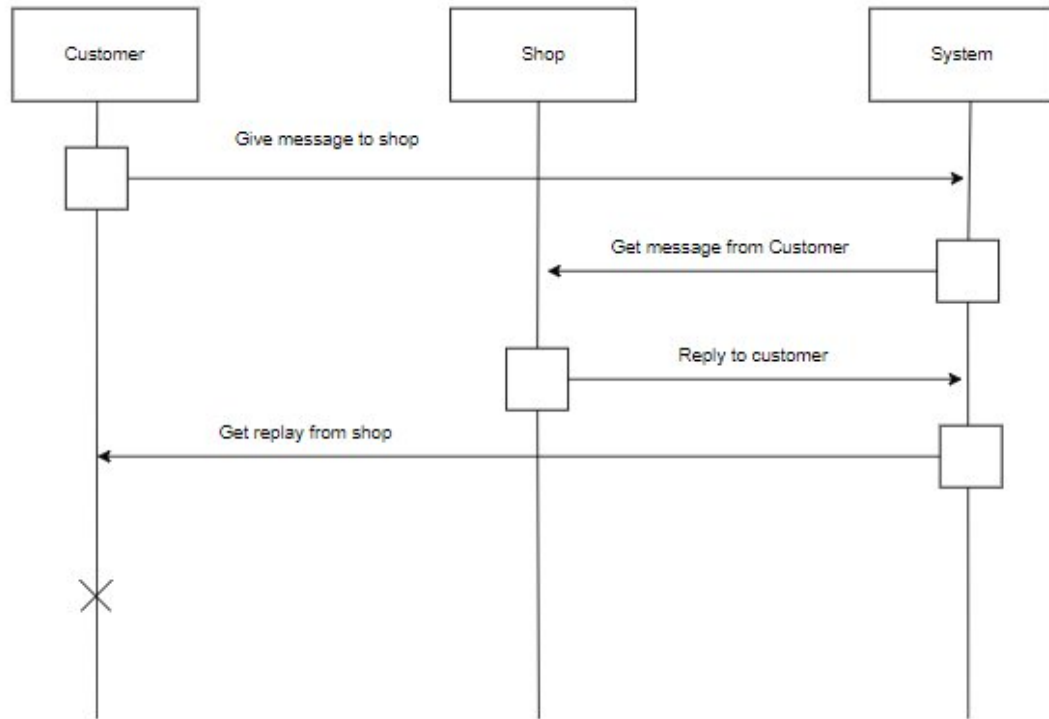


Figure 6.16 Swim Lane diagram for conversation

Swim lane diagram of Withdraw money

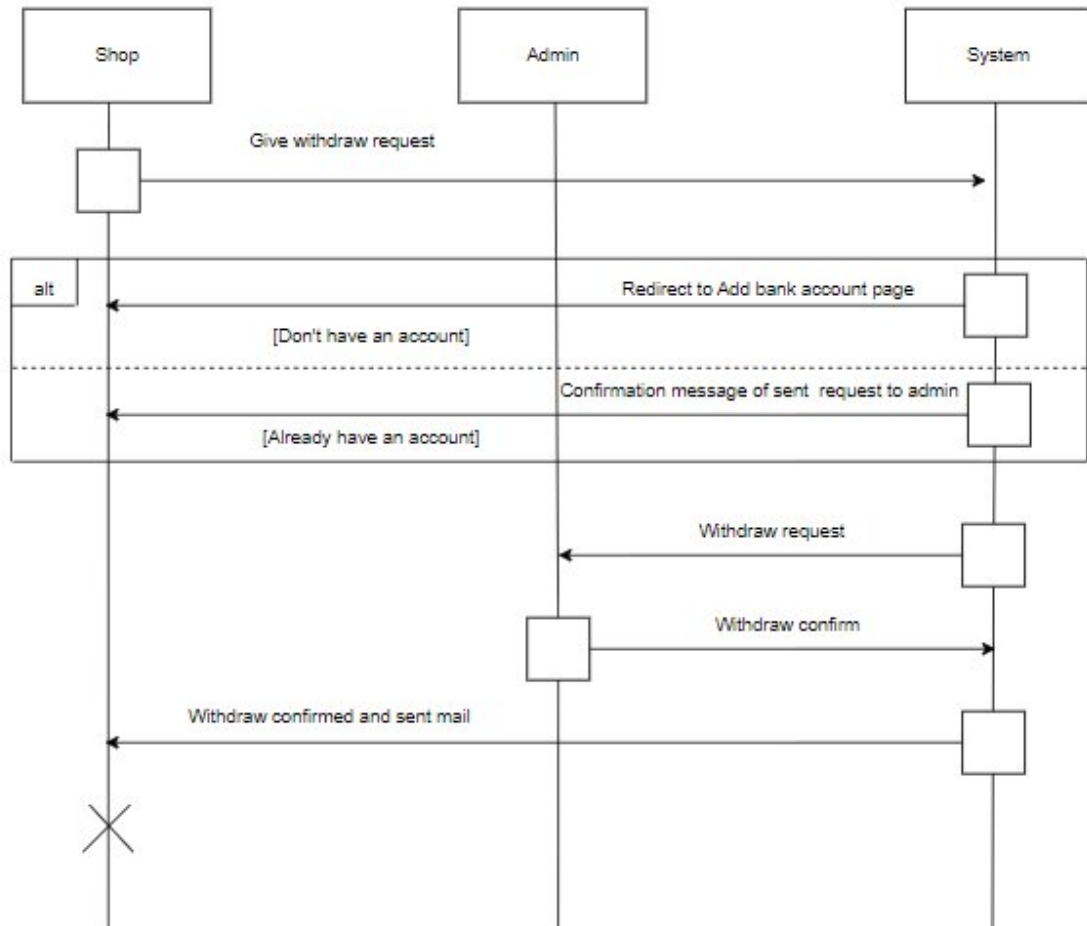


Figure 6.17 Swim Lane diagram for withdraw money

Swim lane diagram of System Monitoring

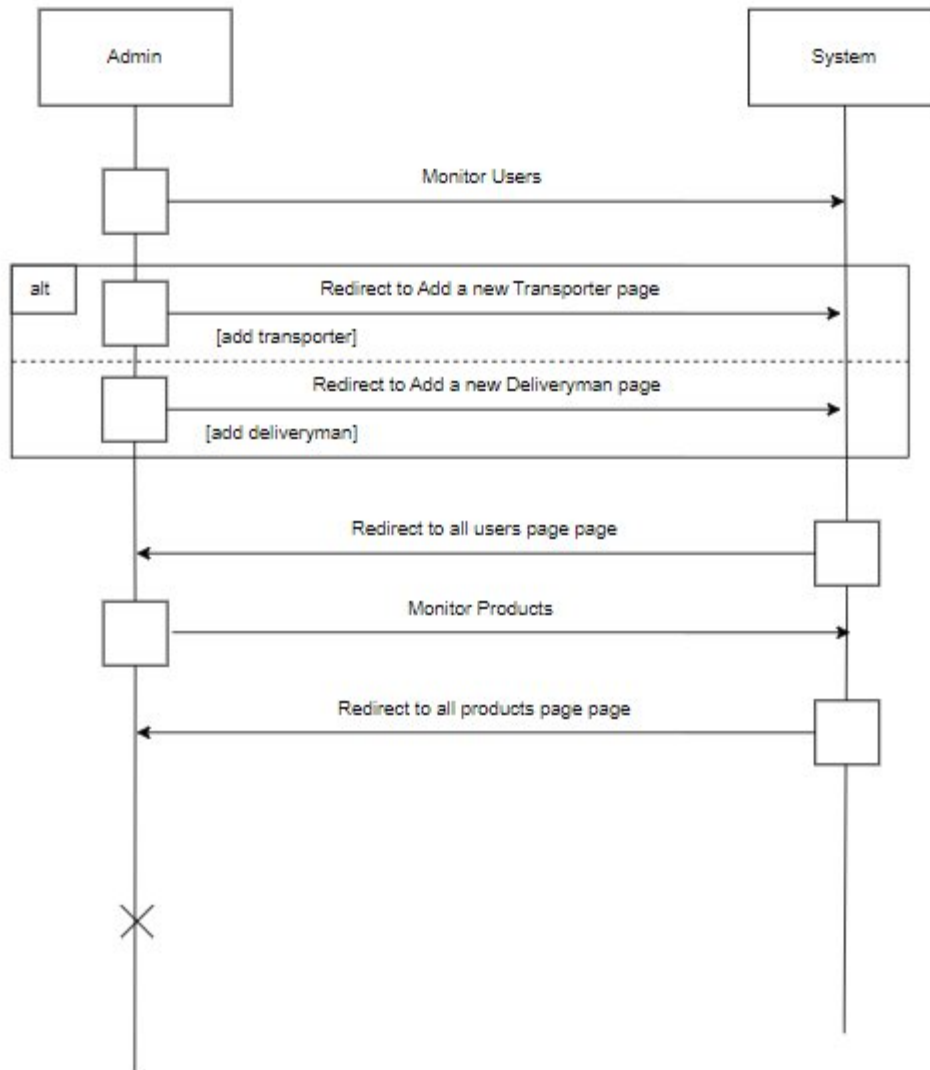


Figure 6.18 Swim Lane diagram for monitor system

6.4 Data Flow Diagram

A data flow diagram (DFD) is a picture that shows how information moves through a system or process. It offers a crystal-clear, organized perspective of how data flows across various system components and interactions.

DFDs employ standardized symbols to depict many elements and their interactions, including rectangles, circles, arrows, and labels. The essential elements of a DFD consist of:

1. **Processes:** Represented by rectangles, processes depict the actions or activities that transform input data into output data.
2. **Data Flows:** Represented by arrows, data flows depict the movement of data from one component to another. They show how data is inputted, processed, and outputted within the system.
3. **Data Stores:** Represented by open-ended rectangles, data stores depict the places where data is stored or retrieved from. They can represent physical storage, databases, or other means of data storage.
4. **External Entities:** Represented by squares or circles, external entities are external sources or destinations of data that interact with the system. They can include users, other systems, or external organizations.

The DFD illustrates the flow of data by showing how information enters and leaves the system, the transformations that occur within processes, and where data is stored. It provides a high-level overview of the system's functionality, allowing stakeholders to understand the interactions between different components and identify potential areas for improvement, data redundancy, or bottlenecks.

DFDs are commonly used in system analysis and design, requirements gathering, and process modeling. They serve as a valuable tool for communication and documentation, helping stakeholders visualize and analyze the flow of information within a system.

Level - 0 (Context Level)

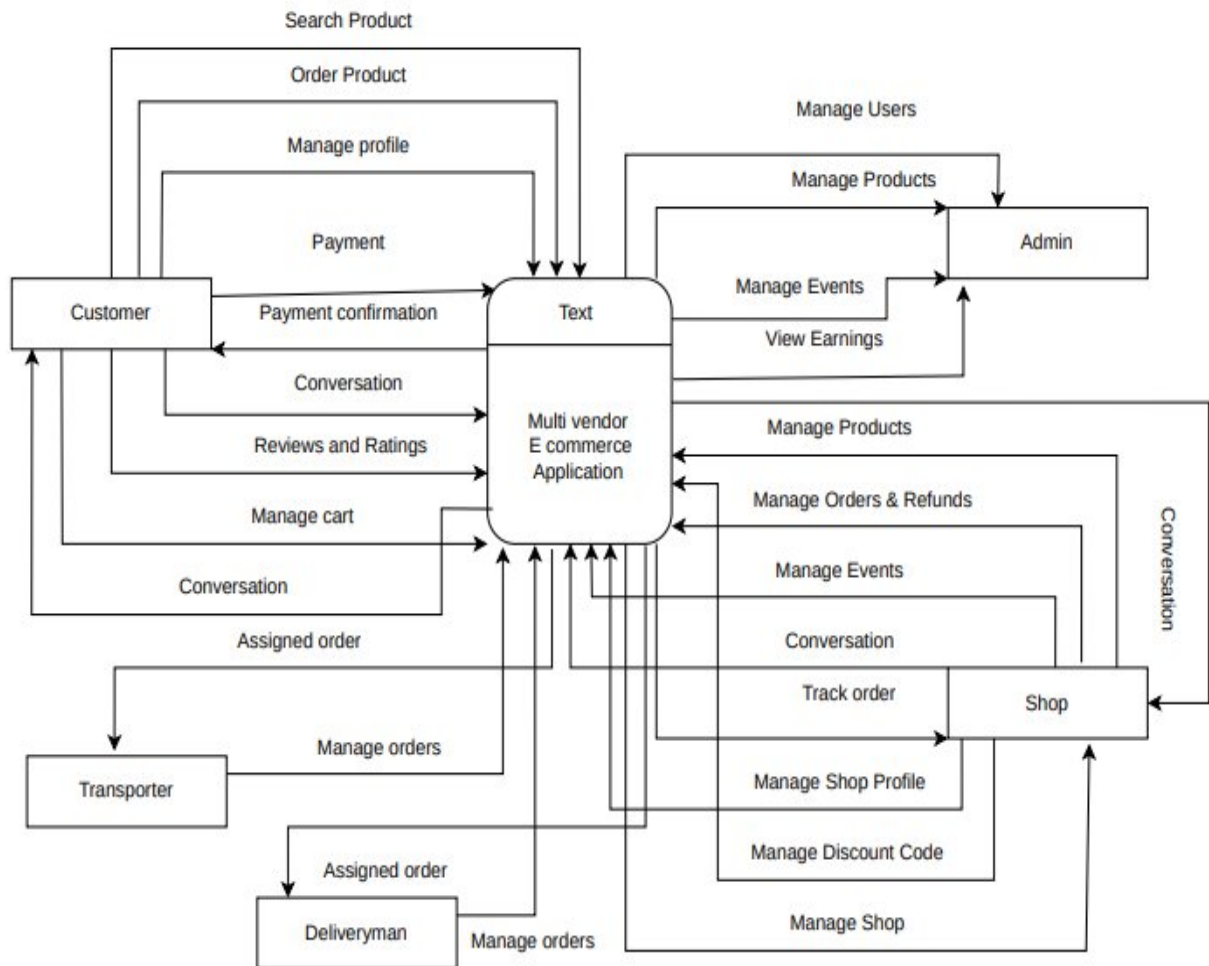


Figure: 6.19 Context Level DFD

Level 1 DFD

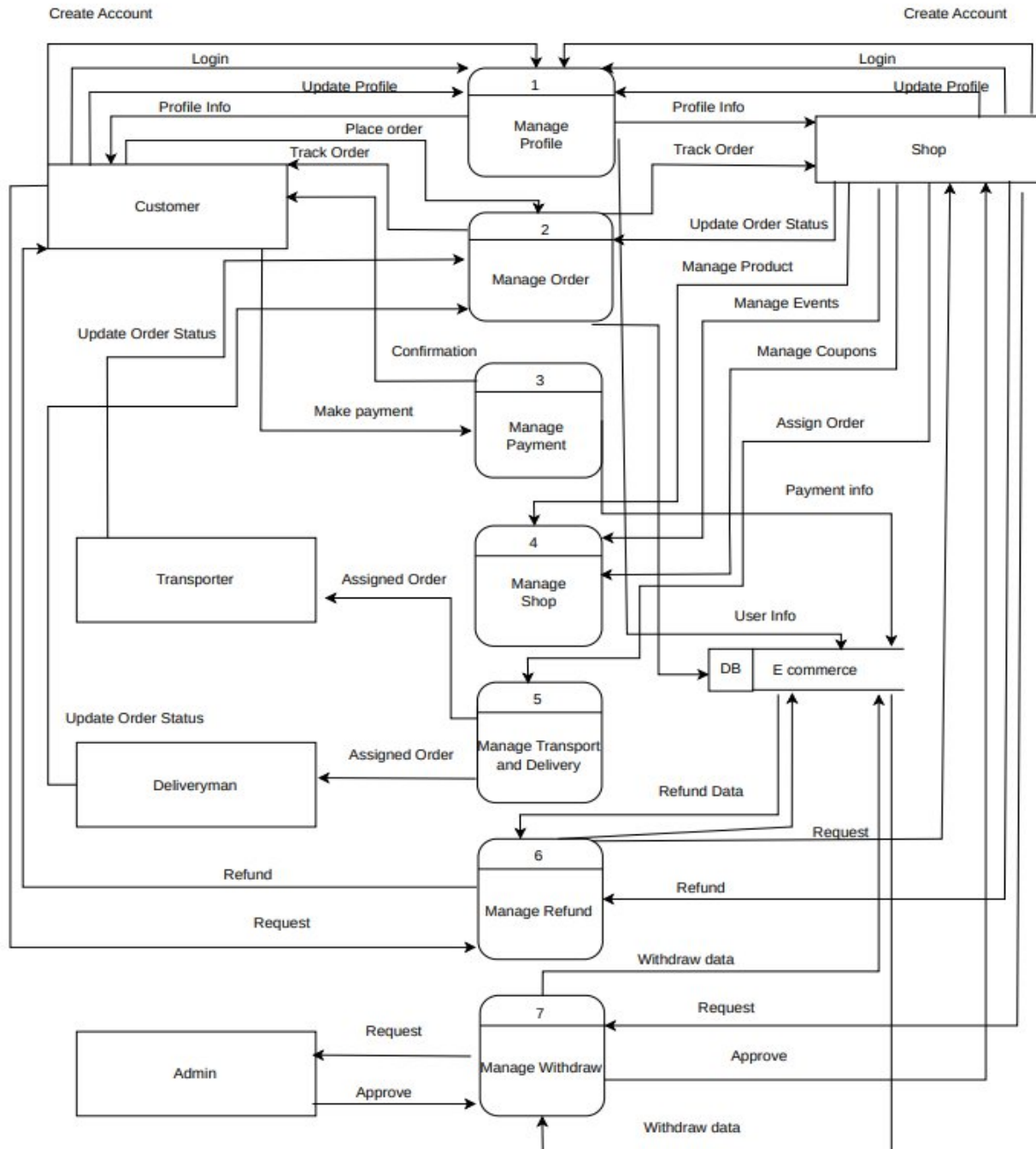


Figure: 6.20 Level 1 DFD

Level 2 DFD of Process 1

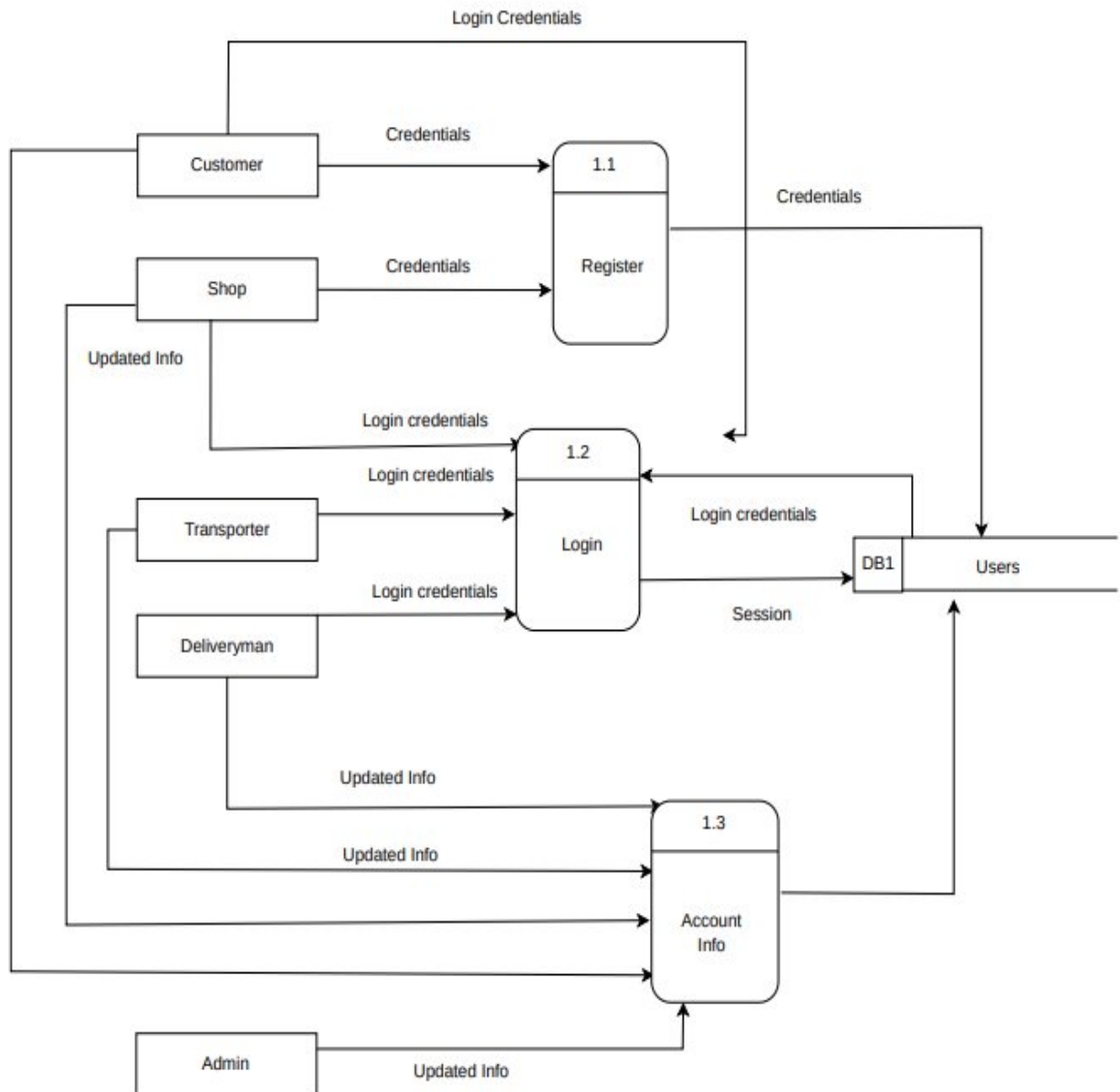


Figure: 6.20 Level 2 DFD of Process 1

Level 2 DFD of Process 2

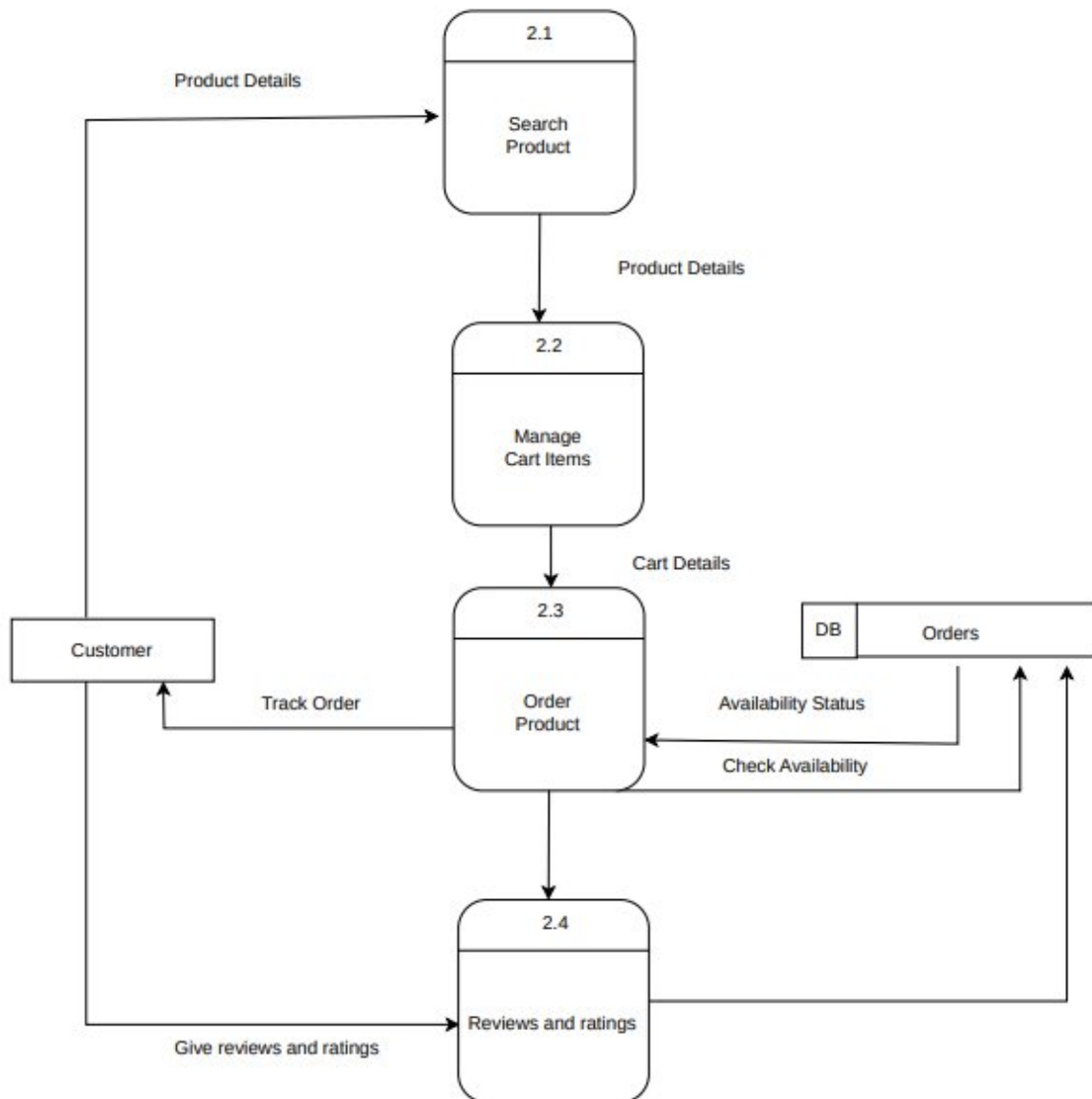


Figure: 6.21 Level 2 DFD of Process 2

Level 2 DFD of Process 3

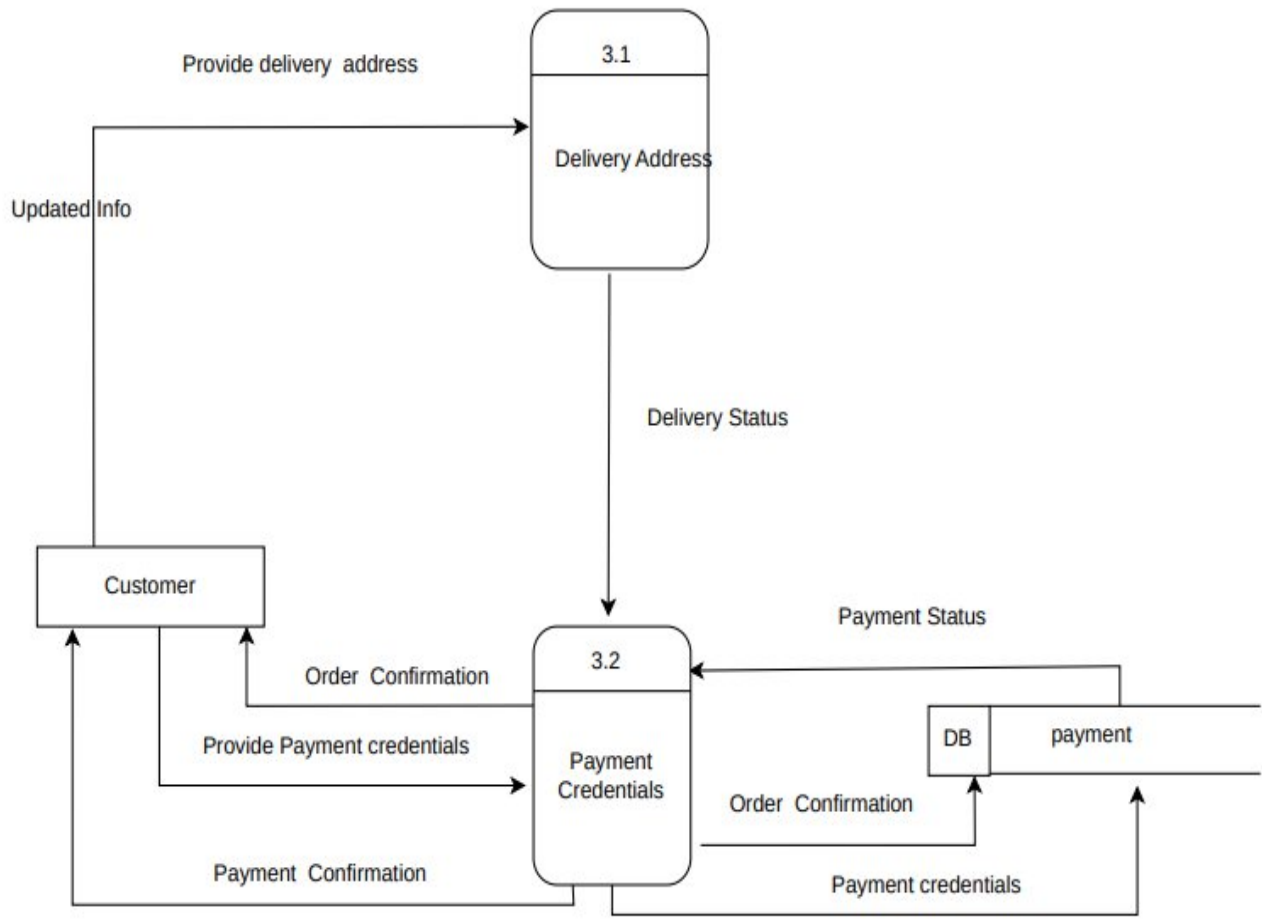


Figure: 6.22 Level 2 DFD of Process 3

Level 2 DFD of Process 4

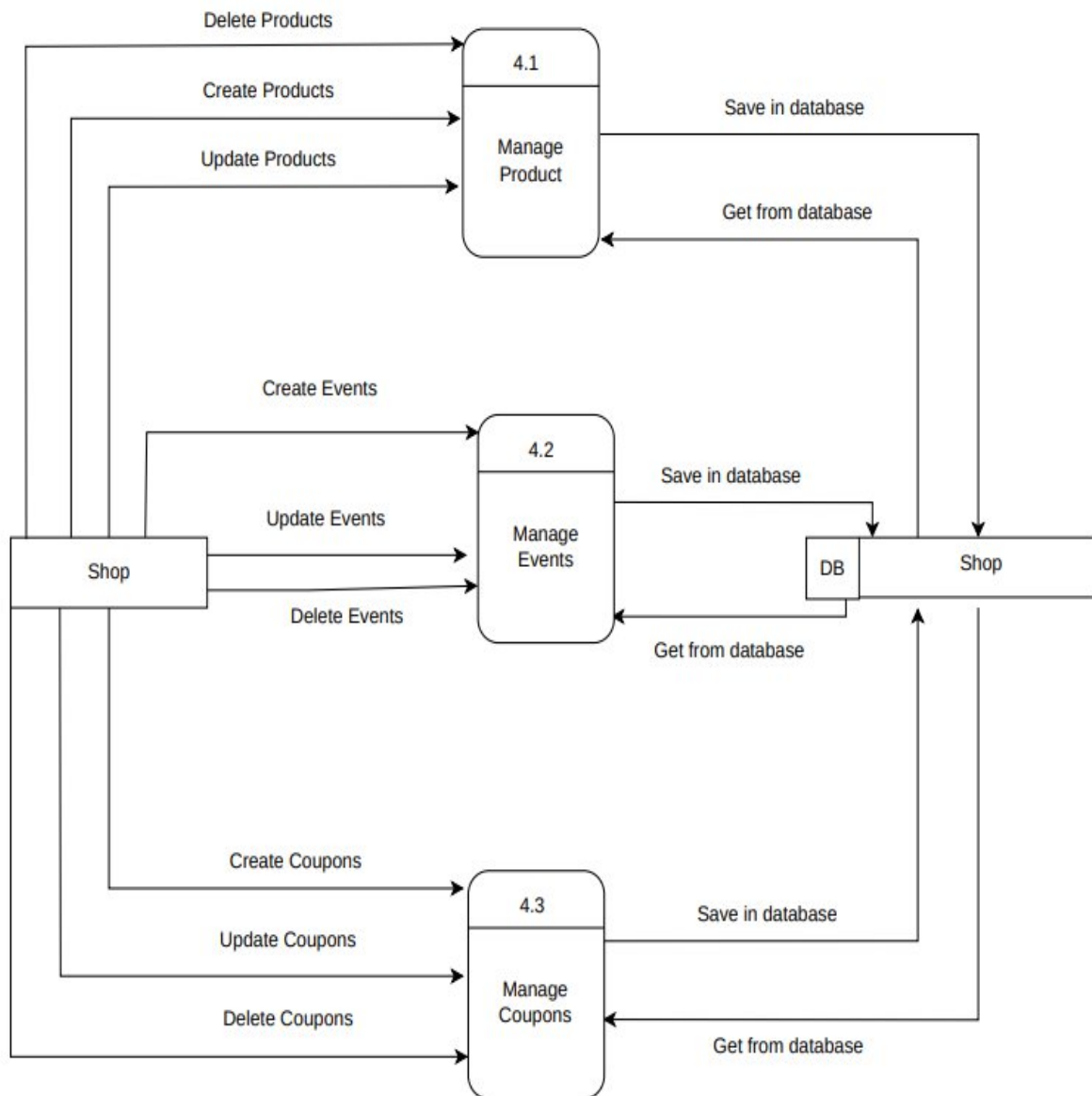


Figure: 6.23 Level 2 DFD of Process 4

Level 2 DFD of Process 5

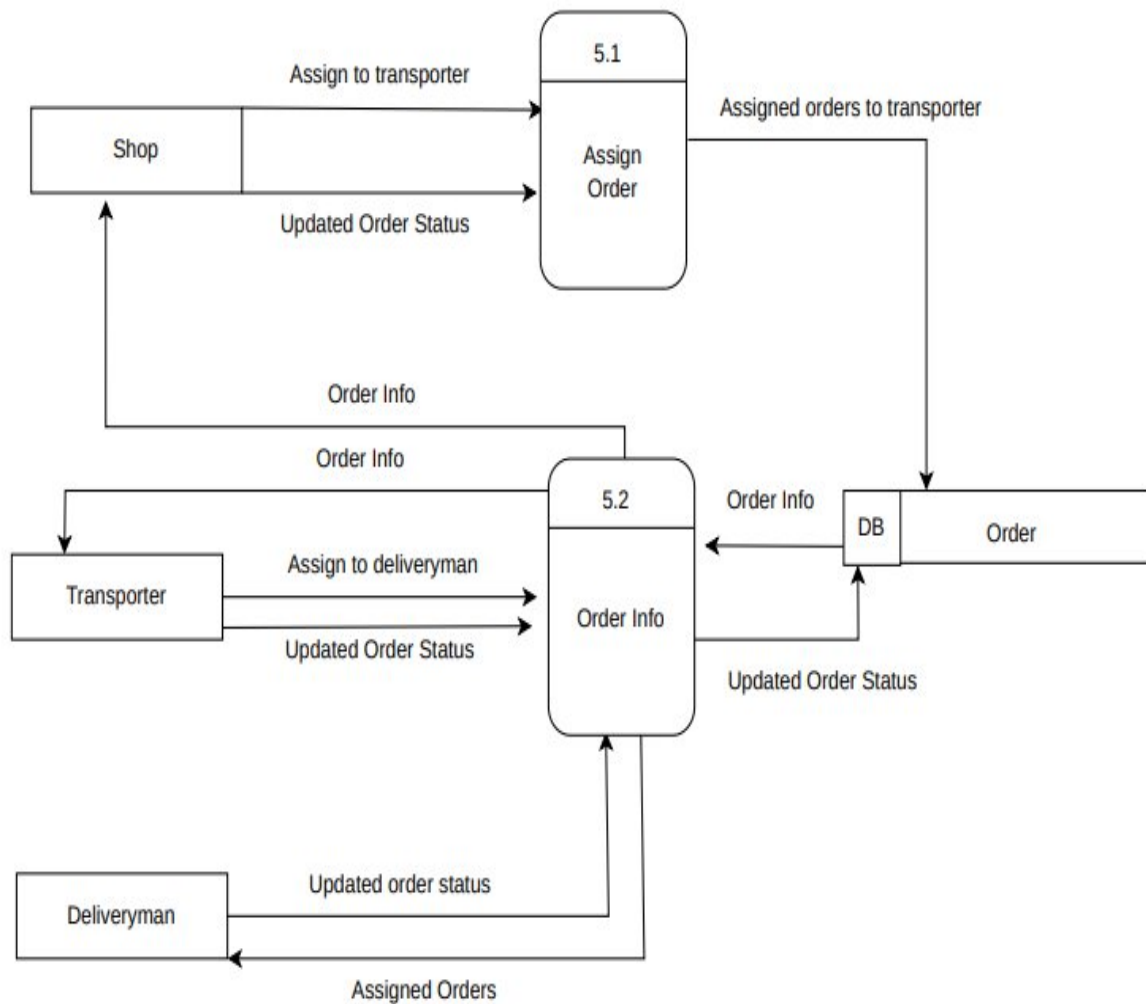


Figure: 6.24 Level 2 DFD of Process 5

Level 2 DFD of Process 6

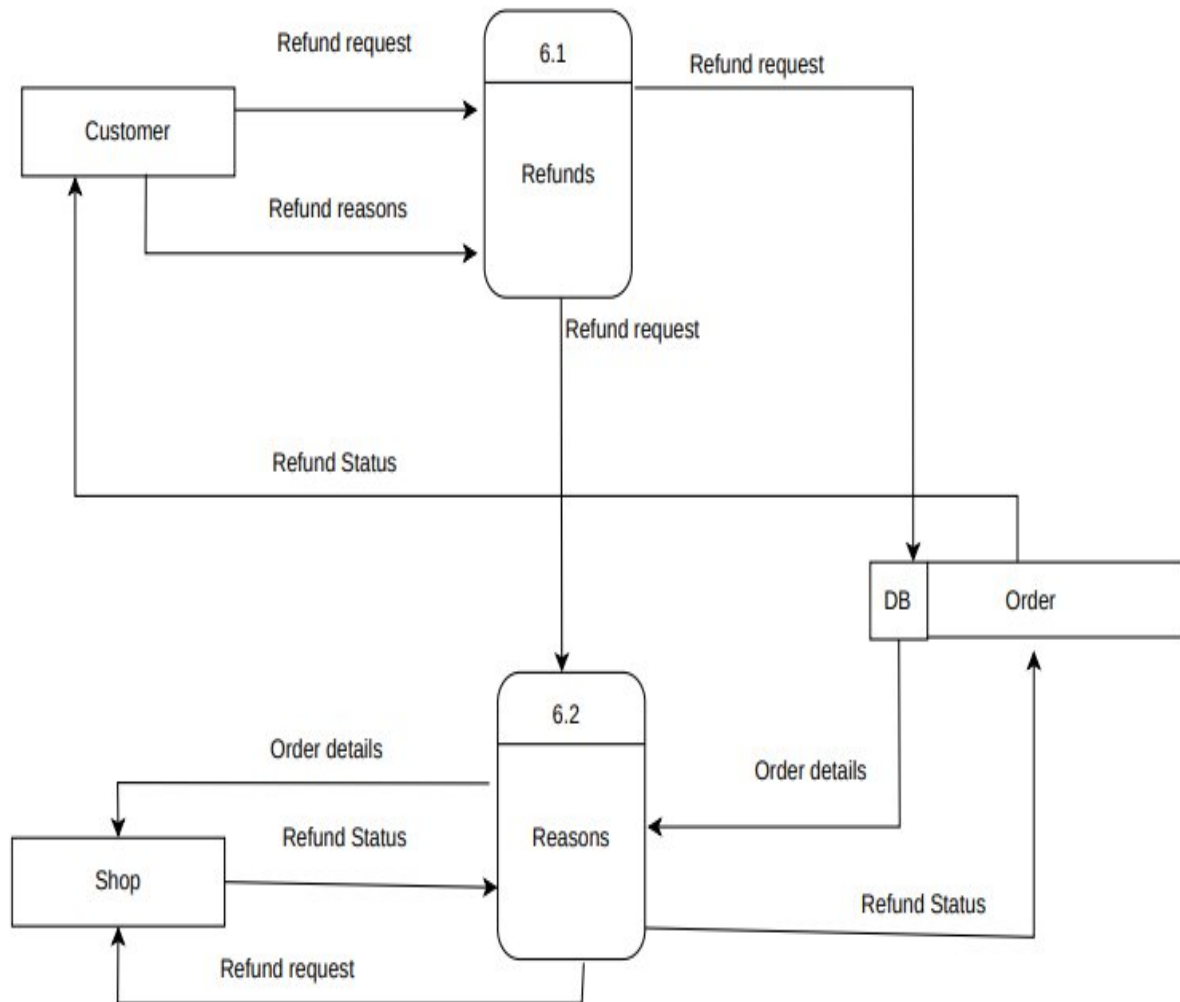


Figure: 6.25 Level 2 DFD of Process 6

Level 2 DFD of Process 7

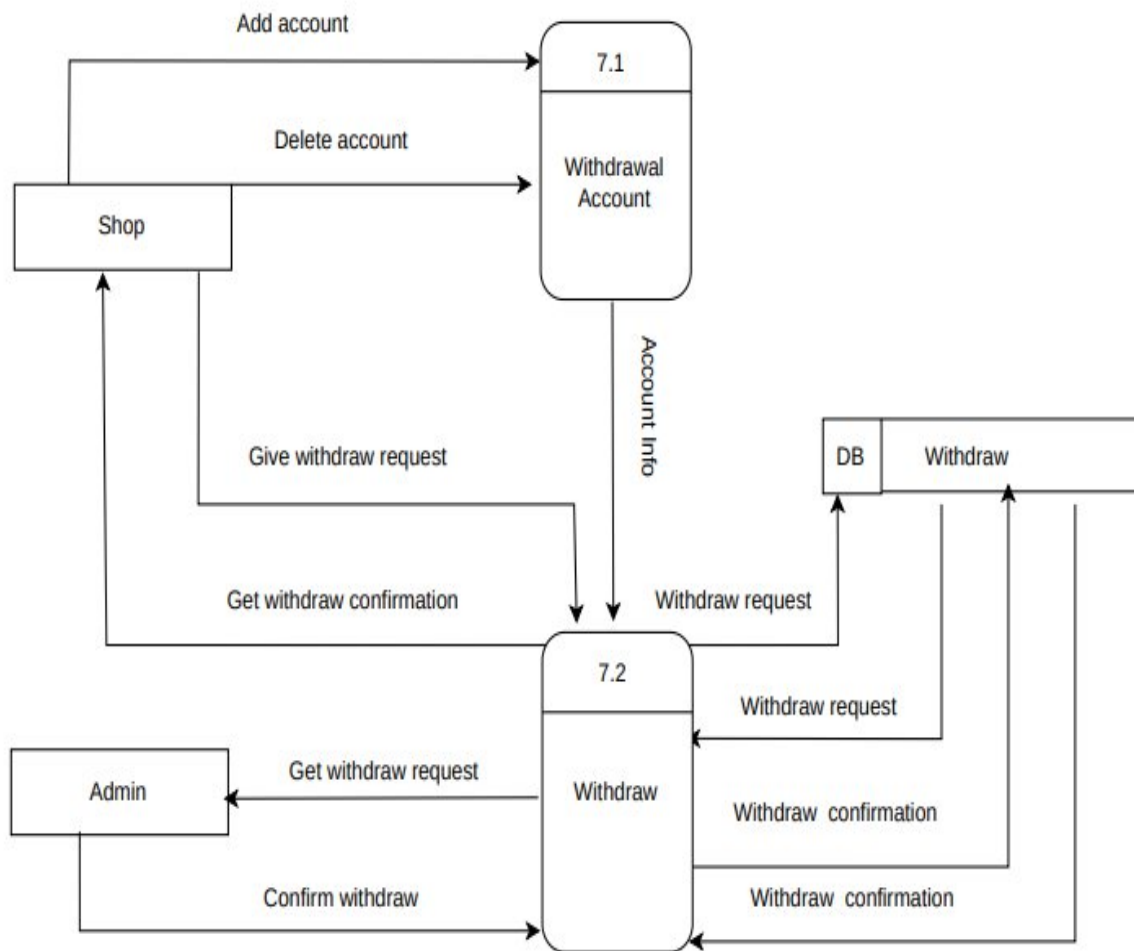


Figure: 6.26 Level 2 DFD of Process 7

Entity Relationship Diagram

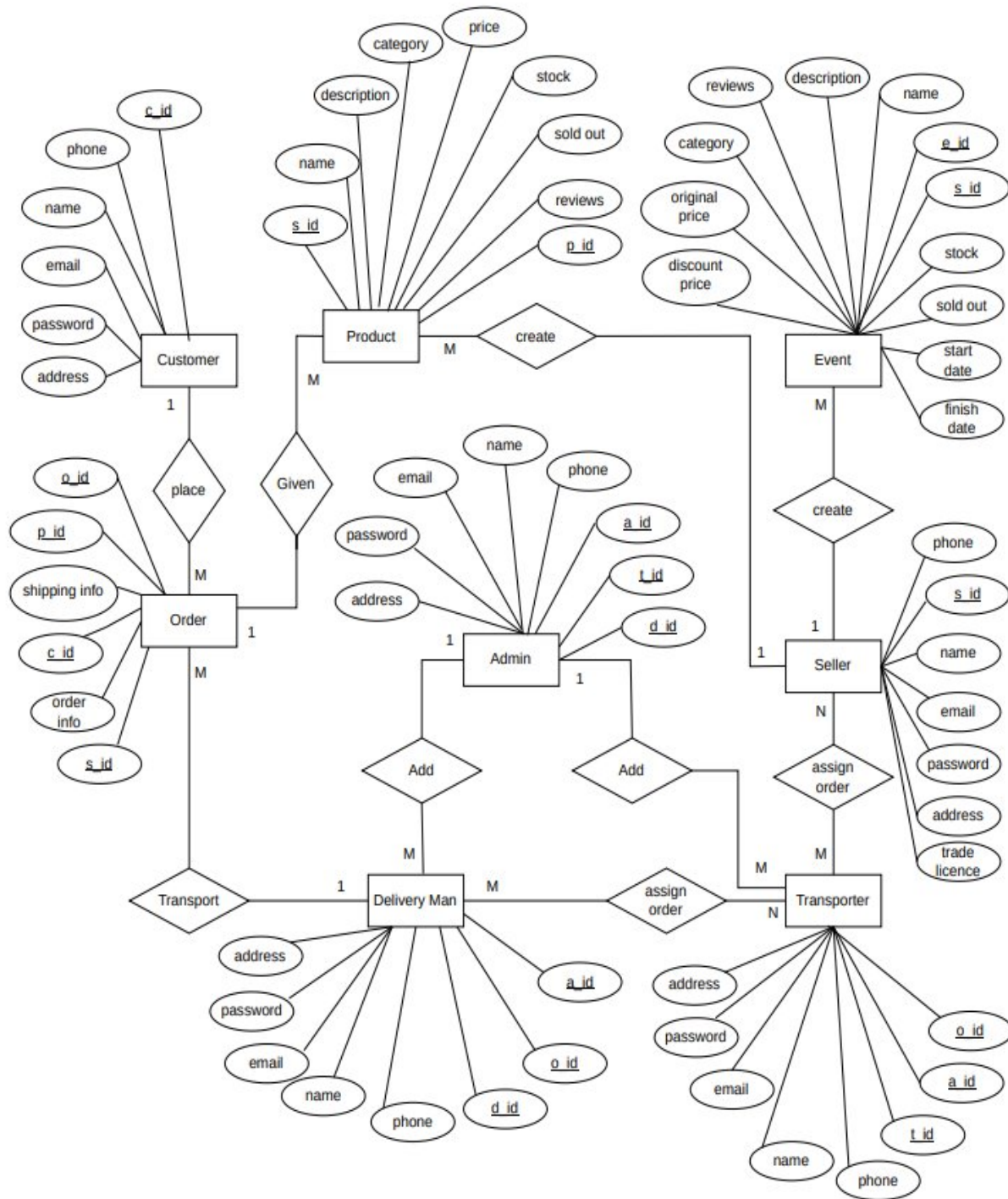


Figure: 6.27 Entity Relationship Diagram

6.5 Database Structure:

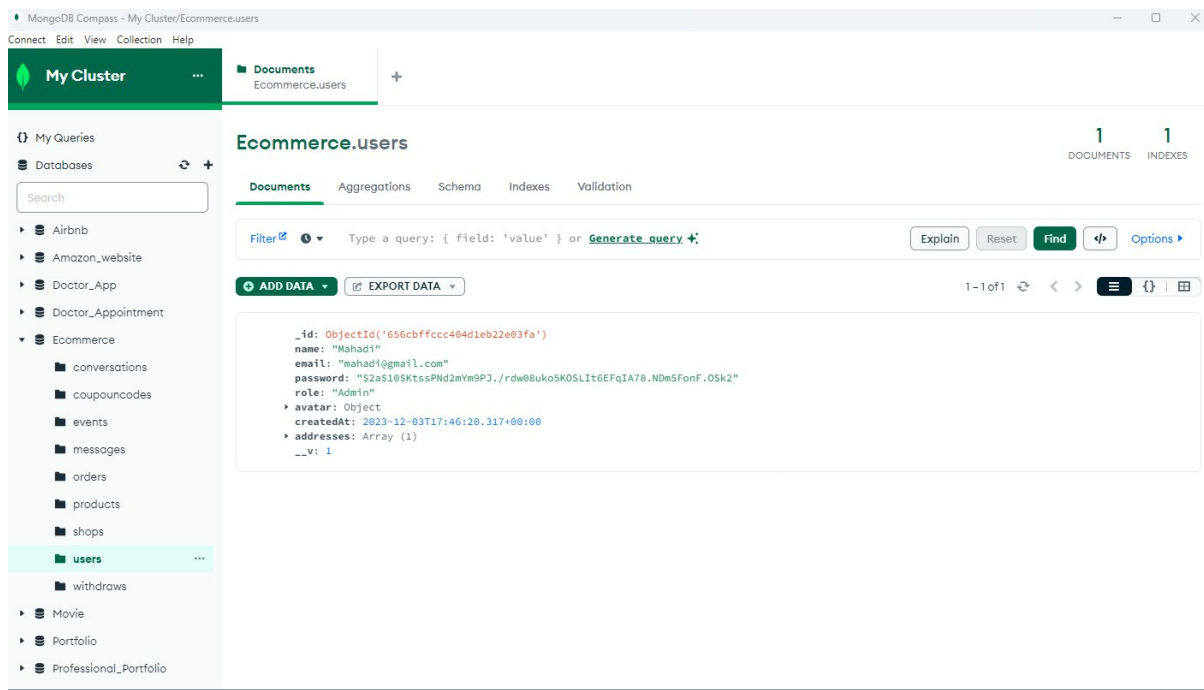


Figure 6.28: User Schema

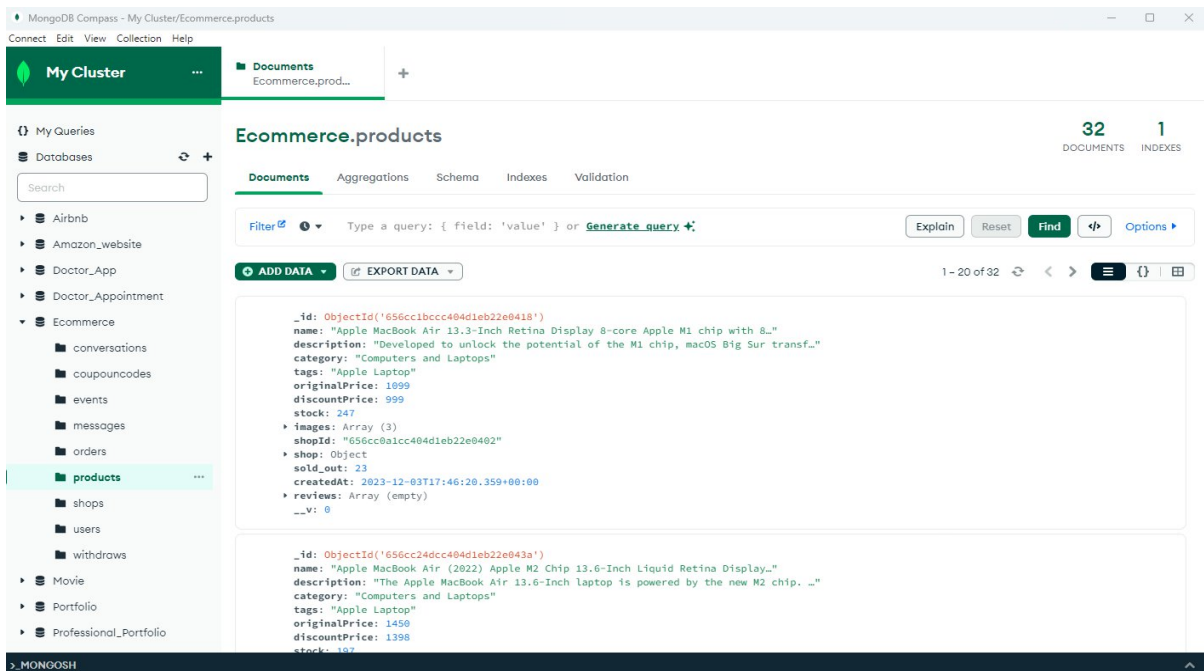


Figure 6.29: Product Schema

Chapter 7

Designing

Login to your account

Email address

Password

☐ Remember me [Forgot your password?](#)

Not have any account? [Sign Up](#)

Figure 7. 1 Login page



[Go Dashboard >](#)

[Home](#) [Best Selling](#) [Products](#) [Events](#) [FAQ](#)

[All Categories](#)

[Profile](#)
[All Orders](#)
[All Refunds](#)
[Inbox](#)
[Track Order](#)
[Change Password](#)
[Address](#)
[Admin Dashboard](#)
[Log out](#)



Full Name
Mahadi

Email Address
mahadi@gmail.com

Phone Number

Enter your password

Figure 7. 2 Customer Dashboard

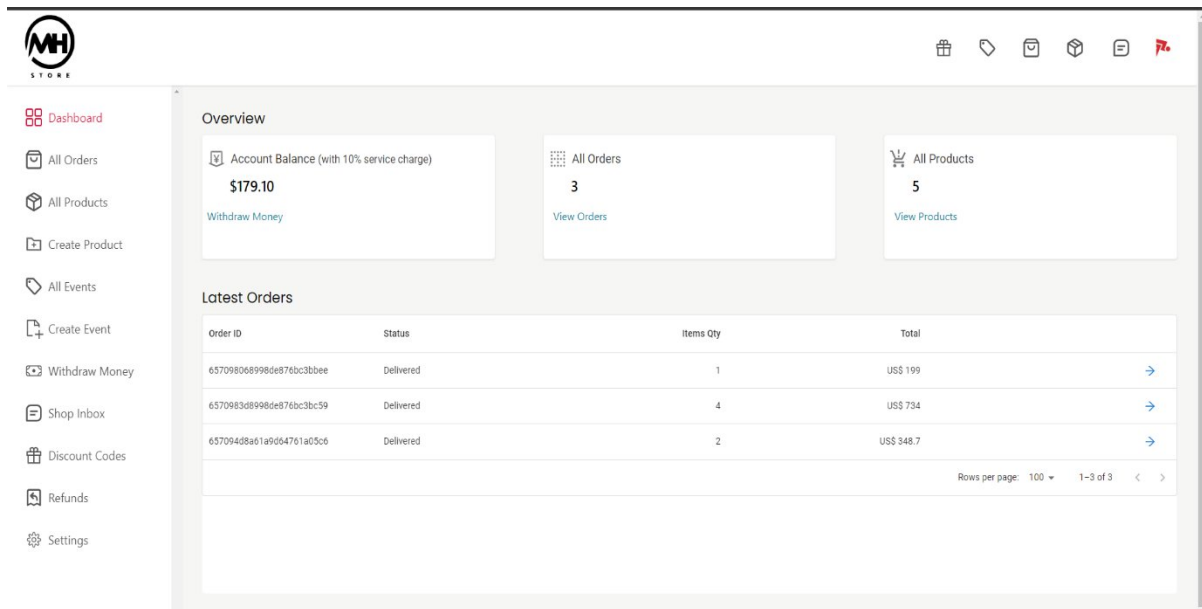


Figure 7. 3 Shop Dashboard

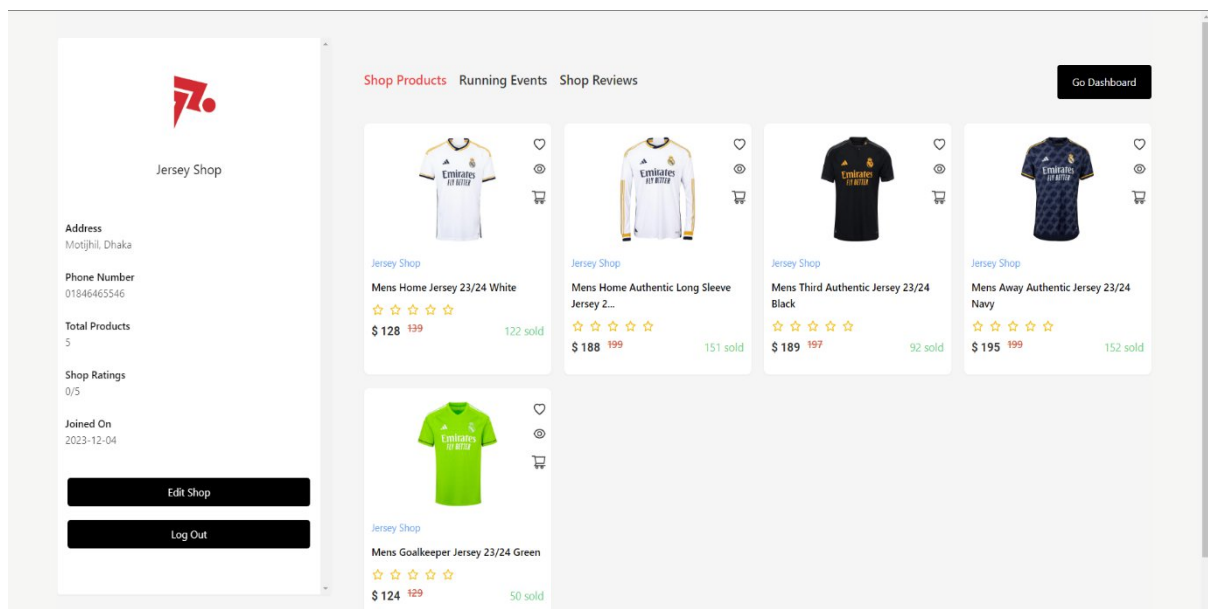


Figure 7. 4 Shop Page

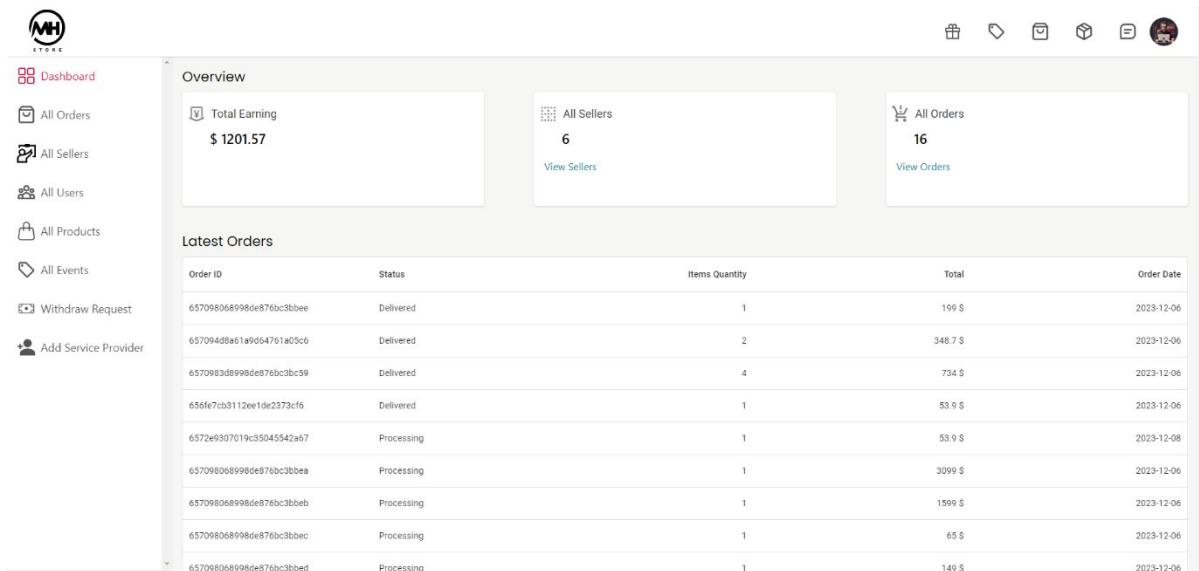


Figure 7. 5 Admin Dashboard

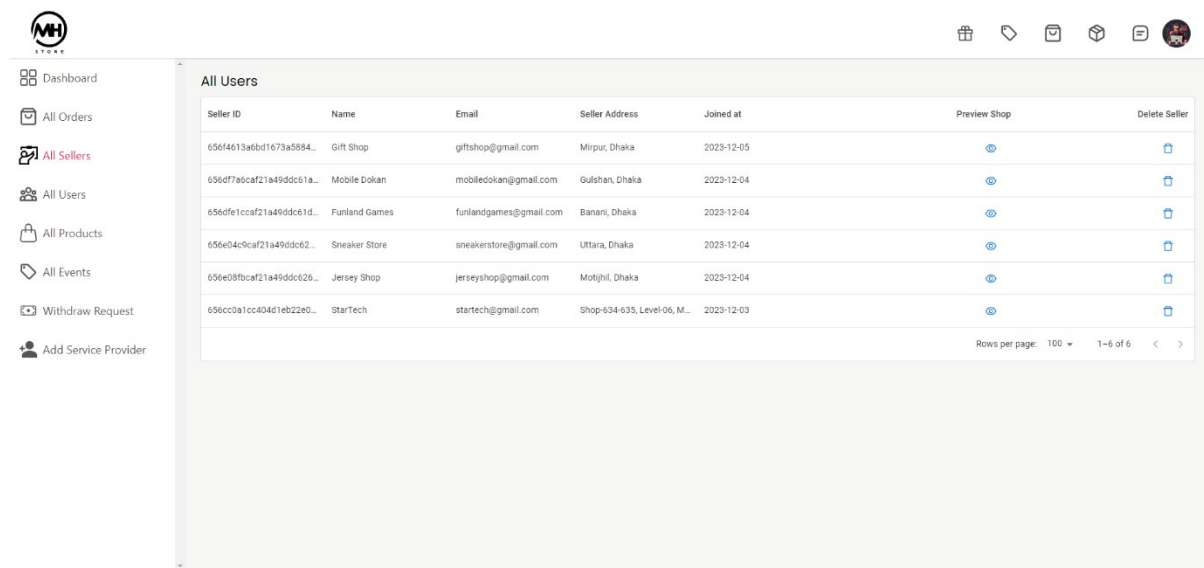


Figure 7. 6 Admin All Seller Monitoring page

Chapter 8

Quality Assurance and Testing

Software testing is a critical step in the creation of software that entails assessing a software component to find any discrepancies between the intended output and the output that the software actually produces. It seeks to evaluate the product's qualities and characteristics to make sure that it satisfies the requirements and performs as planned.

Testing serves as a means of verification and validation:

Verification: It entails determining if the product complies with the requirements established at the start of the development process. It emphasizes verifying that the item operates as planned and adheres to the required standards.

Validation: The product is developed in accordance with the customer's expectations and satisfies the defined standards thanks to this method. It entails assessing the finished item to see if it corresponds with the anticipated usage and functioning.

8.1 System Quality Management

System Quality Management is essential for maintaining a software system's overall quality. It entails putting in place a quality management software system that automates and links every division of a firm that is subject to regulation or that complies with ISQ. A Total Quality Management (TQM) system or Quality Management System (QMS) makes it easier for departments in a company to collaborate and communicate with one another at various stages of product development.

There are two main types of quality that are important in software development:

Internal Quality:

- Test coverage: The extent to which the software has been tested to ensure that all functionality is exercised.
- Testability: The ease with which the software can be tested, including factors like code maintainability and readability.
- Portability: The ability of the software to be easily transferred or adapted to different environments or platforms.

- Thread-safeness: The ability of the software to handle multiple threads or concurrent operations without conflicts or data corruption.
- Conciseness: The clarity and brevity of the code, making it easier to understand and maintain.
- Maintainability: the simplicity with which the program may be updated, mended, or modified during the course of its life.
- Documentation: The presence and quality of documentation that describes the software's functionality, design, and usage.
- Legibility: The readability and clarity of the code, making it easier for developers to understand and work with.

External Quality:

- Features: The functionality and capabilities of the software that meet user needs and requirements.
- Speed: The responsiveness and performance of the software in terms of executing tasks and processing data.
- Space: The efficiency of memory usage and storage requirements of the software.
- Network usage: The efficiency and effectiveness of the software's network communication and data transfer.
- Stability: The software's capacity to function without errors or failures, as well as its dependability and robustness.
- Robustness: The software's ability to handle unexpected or erroneous inputs or conditions without failure.
- Ease-of-use: The user-friendliness and intuitiveness of the software's interface and interaction.

- Determinism: The predictability and consistency of the software's behavior and outputs given the same inputs.
- Back-compatibility: The ability of the software to work with previous versions or interfaces, ensuring smooth transitions for users.

These qualities, both internal and external, contribute to the overall quality of a software product, ensuring that it meets the desired standards, functions correctly, and satisfies user requirements.

Chapter 9

Conclusion

9.1 Preface

In the present era of modern science and communication, the efficient and effective functioning of operational and management processes is crucial for providing better and uninterrupted services. It is a privilege for me to have had the opportunity to work with a team of dedicated and hardworking engineers who have contributed significantly to achieving this goal. I am truly grateful and extend my heartfelt thanks and appreciation to these remarkable individuals. Their commitment and camaraderie have made a profound impact, and I am privileged to have been a part of such a team.

9.2 Practicum and Its Value

The main objective of this report is to describe a system that enhances the efficiency and speed of project completion while providing proper guidance to small businesses and customers. With the implementation of this system, Stores can book appointments online, keep track of their appointments, and stay updated on payment status. This system aims to improve user-friendliness and streamline the point of sale process.

By adopting this system, the point of sale system will become more effective and efficient, requiring less time and reducing the need for extensive paperwork compared to traditional methods. The benefits of this system are evident in its user-friendly interface and streamlined processes.

Undergraduate engineering studies provide students with both theoretical and practical knowledge. The practicum program offers an opportunity to apply this knowledge and gain practical working skills by observing real-world operational systems. Our practicum experience was a successful event as it deepened our understanding of the subjects and enhanced our practical skills.

Practical work experience holds immense value as it provides a unique perspective that cannot be obtained through theoretical learning alone. Prior to entering the job market, students should strive to gain real-world work experience in their field of study. Job opportunities are often influenced by the quality of work experiences. Students with better work experiences tend to have more favorable job prospects.

I got the chance to use the information, approaches, resources, and skills I had obtained while pursuing my studies at IUBAT while working as an intern at CLOUDLY INFOTECH LTD..... Standardized development procedures, sound theoretical foundations, and the developer's creativity are all necessary for successful software development.

Overall, the practicum experience and practical work exposure contribute significantly to personal and professional growth, providing valuable insights and enhancing employability in the job market.

Future Plan

As the project is in its initial stage, there are plans to enhance the system by incorporating additional features based on user feedback and requirements. The goal is to continuously improve and maintain the system while providing ongoing support to users. Some of the potential future features that can be considered are:

- Product review system by admin
- Barcode system
- Products return system including giving points
- Shop ranking

9.3 Conclusion

In summary, the Multi-Vendor E-commerce Platform stands as a comprehensive solution, harnessing the prowess of the MERN stack to create a dynamic, user-centric marketplace. It harmonizes the interactions among customers, vendors, transporters, delivery personnel, and administrators, fostering seamless transactions and efficient management. Customers benefit from a wealth of functionalities, vendors wield powerful tools to manage their shops, while transporters and delivery personnel ensure timely order fulfillment. With the system administrator overseeing the platform's integrity and security, this ecosystem thrives as a unified digital marketplace, promoting effective communication and facilitating the ever-evolving landscape of e-commerce.

Chapter 10

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